QUT Centre for Robotics



ENN583 – Foundations of Robotic Vision

Week 9 – Sem 2 – 2026

Prof Niko Suenderhauf

Welcome to Week 9!



CRICOS No.00213J

Learned Geometry and 3D Representations

- Monocular depth
- From Depth to Learned Multi-View Geometry
- Reminder: Classical 3D Representations
- Learned Implicit Representations (NeRFs, SDFs)
- Learned Explicit 3D Representations (3D Gaussian Splatting)

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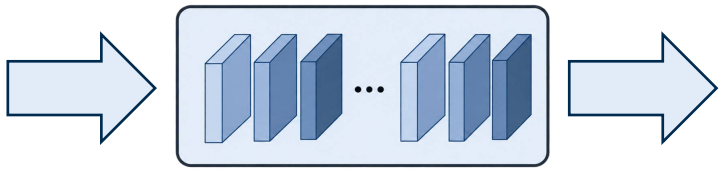
Monocular Depth Prediction

Monocular Depth Prediction

Depth map: for each pixel, the Z-coordinate of the corresponding 3D point in the camera coordinate system.

RGB Image





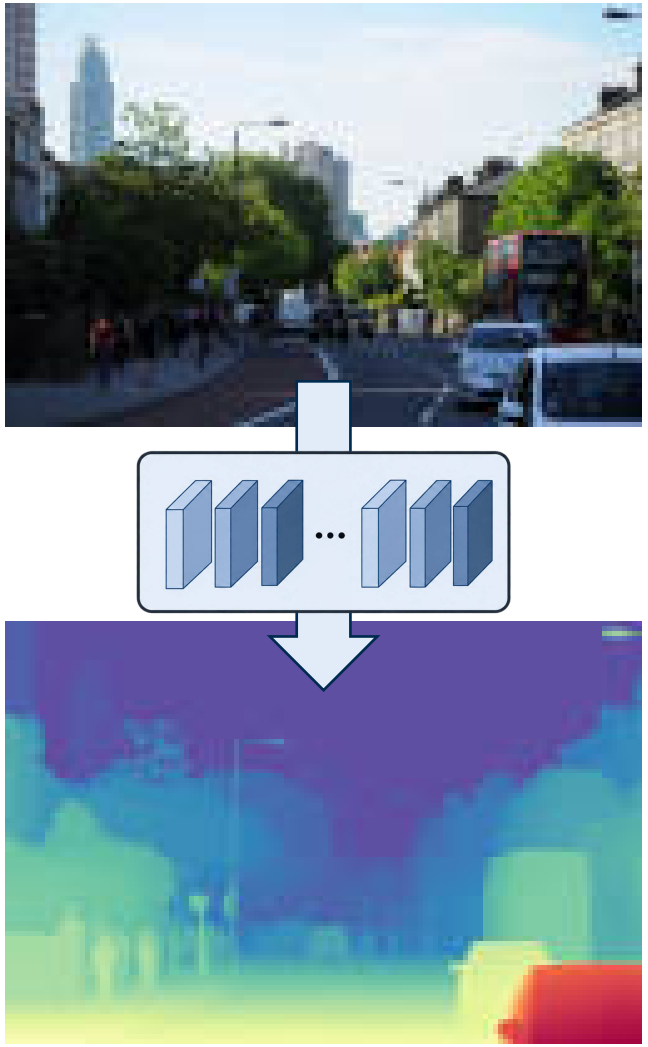


What supervision signal could we use to train a depth prediction network?

Monocular Depth Prediction

What supervision signal could we use to train depth prediction?

- **Direct Supervision**: ground-truth depth from LiDAR, RGB-D, structured light, stereo reconstruction, or synthetic rendering.
- **Self-supervision from geometry:** use stereo pairs or monocular video plus camera motion; predict depth, warp one view into another, and minimise photometric reconstruction error.
- **Pseudo-label / teacher supervision:** use a stronger pretrained depth model to generate targets for large image collections, then train a student model on those predictions.



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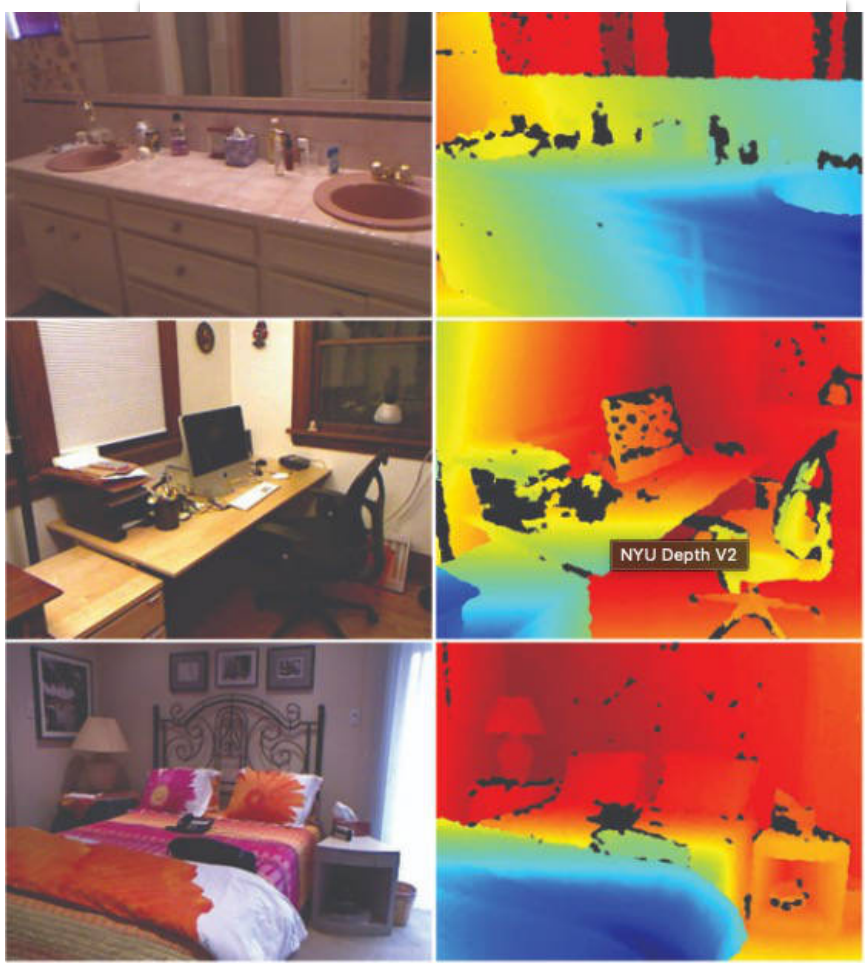
Monocular Depth with Direct Supervision

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Monocular Depth with Direct Supervision

Simple Idea:

- Use a dataset with ground truth depth (e.g. from a RGB-D camera or Lidar),
- Predict the depth values from the image and compare against the ground truth.
- Metric supervision means the model can learn absolute depth, although losses may also emphasise relative scene structure
- Key limitation: collecting dense, accurate ground-truth depth at scale is expensive.



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Monocular Depth with Direct Supervision

*"… the task is inherently ambiguous, and a technically ill-posed problem: **Given an image, an infinite number of possible world scenes may have produced it**. Of course, most of these are physically implausible for real-world spaces, and thus the depth may still be predicted with considerable accuracy.*

*At least one major ambiguity remains, though: the **global scale**. Although extreme cases (such as a normal room versus a dollhouse) do not exist in the data, moderate variations in room and furniture sizes are present.*

*We address this **using a scale-invariant error in addition to more common scale-dependent errors**. This focuses attention on the spatial relations within a scene rather than general scale, and is particularly apt for applications such as 3D modeling, where the model is often rescaled during postprocessing."*



arXiv:1406.2283v1 [cs.CV] 9 Jun 2014

Depth Map Prediction from a Single Image using a Multi-Scale Deep Network

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Christian Puhrsch[1] cpuhrsch@nyu.edu
Rob Fergus[1,2] fergus@cs.nyu.edu

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[2]Facebook AI Research

Abstract

Predicting depth is an essential component in understanding the 3D geometry of a scene. While for stereo images local correspondence suffices for estimation, finding depth relations from a *single image* is less straightforward, requiring integration of both global and local information from various cues. Moreover, the task is inherently ambiguous, with a large source of uncertainty coming from the overall scale. In this paper, we present a new method that addresses this task by employing two deep network stacks: one that makes a coarse global prediction based on the entire image, and another that refines this prediction locally. We also apply a scale-invariant error to help measure depth relations rather than scale. By leveraging the raw datasets as large sources of training data, our method achieves state-of-the-art results on both NYU Depth and KITTI, and matches detailed depth boundaries without the need for superpixelation.

1 Introduction

Estimating depth is an important component of understanding geometric relations within a scene. In turn, such relations help provide richer representations of objects and their environment, often leading to improvements in existing recognition tasks [18], as well as enabling many further applications such as 3D modeling [16, 6], physics and support models [18], robotics [4, 14], and potentially reasoning about occlusions.

While there is much prior work on estimating depth based on stereo images or motion [17], there has been relatively little on estimating depth from a *single* image. Yet the monocular case often arises in practice: Potential applications include better understandings of the many images distributed on the web and social media outlets, real estate listings, and shopping sites. These include many examples of both indoor and outdoor scenes.

There are likely several reasons why the monocular case has not yet been tackled to the same degree as the stereo one. Provided accurate image correspondences, depth can be recovered deterministically in the stereo case [5]. Thus, stereo depth estimation can be reduced to developing robust image point correspondences — which can often be found using local appearance features. By contrast, estimating depth from a single image requires the use of monocular depth cues such as line angles and perspective, object sizes, image position, and atmospheric effects. Furthermore, a global view of the scene may be needed to relate these effectively, whereas local disparity is sufficient for stereo.

Moreover, the task is inherently ambiguous, and a technically ill-posed problem: Given an image, an infinite number of possible world scenes may have produced it. Of course, most of these are physically implausible for real-world spaces, and thus the depth may still be predicted with considerable accuracy. At least one major ambiguity remains, though: the global scale. Although extreme cases (such as a normal room versus a dollhouse) do not exist in the data, moderate variations in room

1

Monocular Depth with Direct Supervision

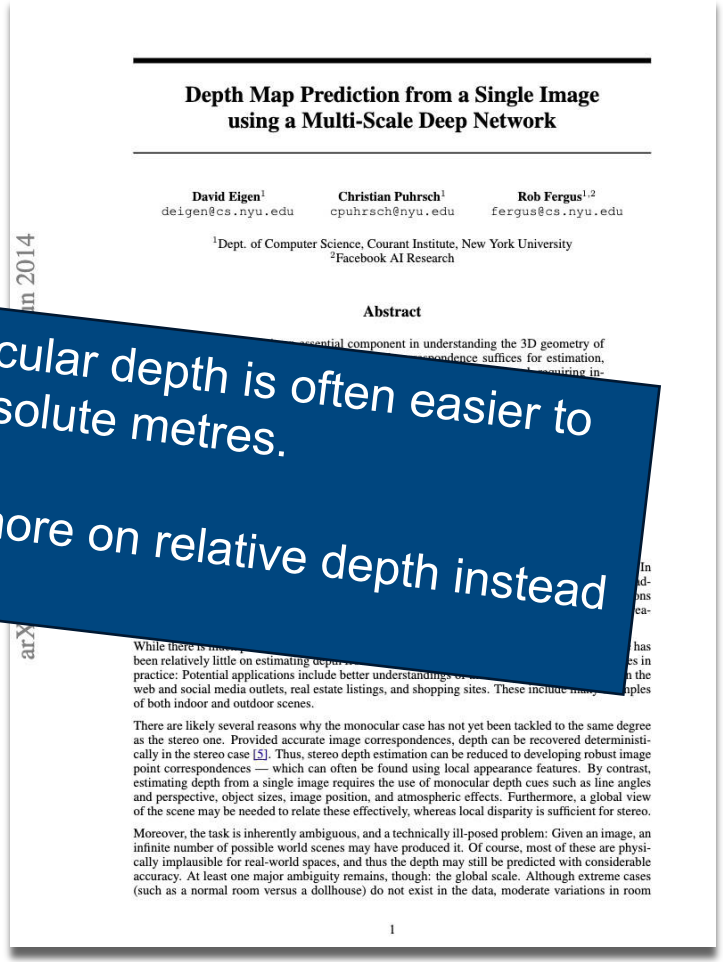
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Even with metric supervision, monocular depth is often easier to learn up to a global scale than in absolute metres.

This sets up later trends that focus more on relative depth instead of metric depth prediction.



Depth Map Prediction from a Single Image using a Multi-Scale Deep Network

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arXiv … 2014

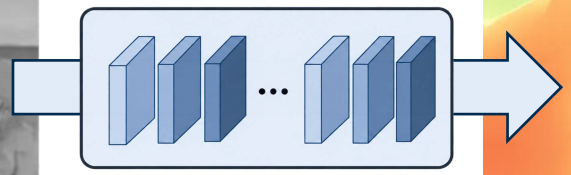
Monocular Depth from Self-Supervision and Multi-View Geometry

Monocular Depth from Self-Supervision and Multi-View Geometry

- Assume we have a rectified stereo camera
- A network predicts disparity from the left image.
- How can we use that to form a loss to train the network?

Left Image





Predicted Disparity



Right Image

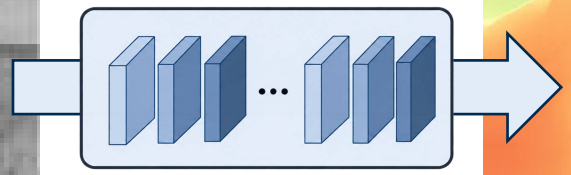


Monocular Depth from Self-Supervision and Multi-View Geometry

- For every pixel in the left image, the disparity tells us the corresponding pixel in the right image!
- If $\hat{d} = D(u, v)$ is the disparity at (u, v) then the corresponding pixel $(u', v') = (u-\hat{d}, v)$!
- → Set up a photometric loss, compare the pixel intensities: $I_{\text{left}}(u, v) - I_{\text{right}}(u', v')$

Left Image





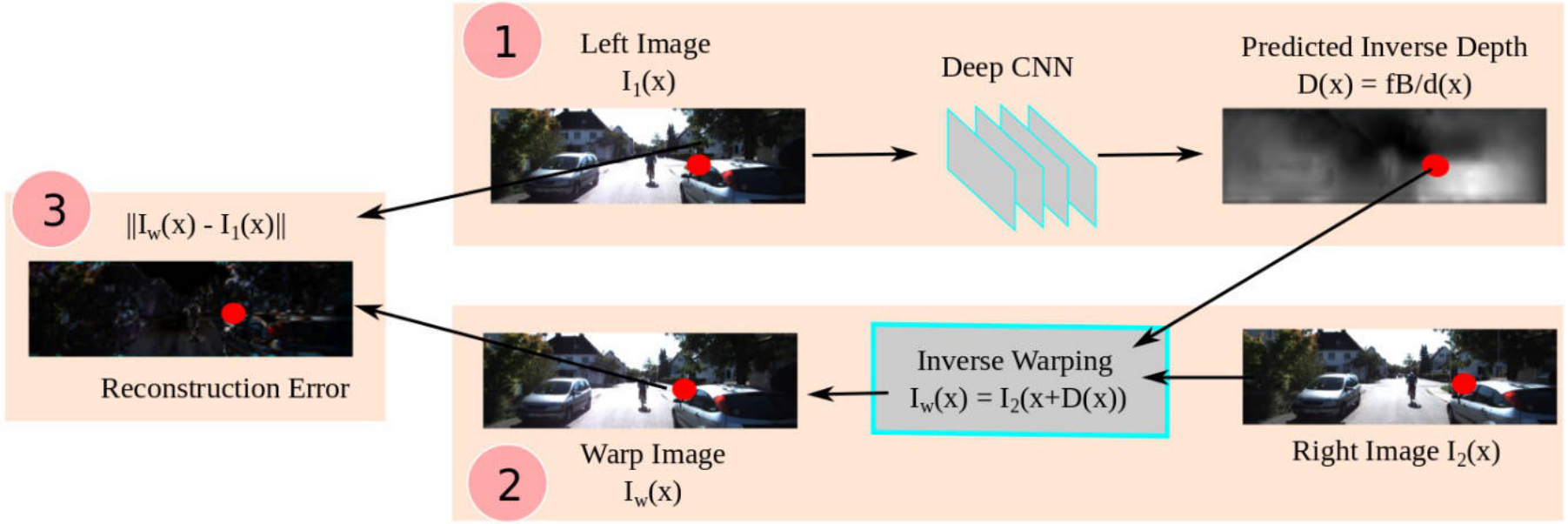
Predicted Disparity



Right Image



Monocular Depth from Self-Supervision and Multi-View Geometry



arXiv:1603.04992v2 [cs.CV] 29 Jul 2016

Unsupervised CNN for Single View Depth Estimation: Geometry to the Rescue

Ravi Garg, Vijay Kumar B G, Gustavo Carneiro, Ian Reid

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{ravi.garg,vijay.kumar,gustavo.carneiro,ian.reid}@adelaide.edu.au

Abstract. A significant weakness of most current deep Convolutional Neural Networks is the need to train them using vast amounts of manually labelled data. In this work we propose a unsupervised framework to learn a deep convolutional neural network for single view depth prediction, without requiring a pre-training stage or annotated ground-truth depths. We achieve this by training the network in a manner analogous to an autoencoder. At training time we consider a pair of images, source and target, with small, known camera motion between the two such as a stereo pair. We train the convolutional encoder for the task of predicting the depth map for the source image. To do so, we explicitly generate an inverse warp of the target image using the predicted depth and known inter-view displacement, to reconstruct the source image; the photometric error in the reconstruction is the reconstruction loss for the encoder. The acquisition of this training data is considerably simpler than for equivalent systems, requiring no manual annotation, nor calibration of depth sensor to camera. We show that our network trained on less than half of the KITTI dataset gives comparable performance to that of the state-of-the-art supervised methods for single view depth estimation. [1]

1 Introduction

The availability of very large human annotated datasets like Imagenet [6] has led to a surge of deep learning approaches successfully addressing various vision problems. Trained initially on tasks such as image classification, and fine-tuned to fit other tasks, supervised CNNs are now state-of-the-art for object detection [14], per-pixel image classification [28], depth and normal prediction from single image [22], human pose estimation [9] and many other applications. A significant and abiding weakness, however, is the need to accrue labeled data for the supervised learning. Providing per-pixel segmentation masks on large datasets like CoCo [23], or classification labels for Imagenet requires significant human effort and is prone to error. Supervised training for single view depth estimation for outdoor scenes requires expensive hardware and careful acquisition[29,8,24,21].

[1] Find the model and other imformation on the project github page: `https://github.com/Ravi-Garg/Unsupervised_Depth_Estimation`

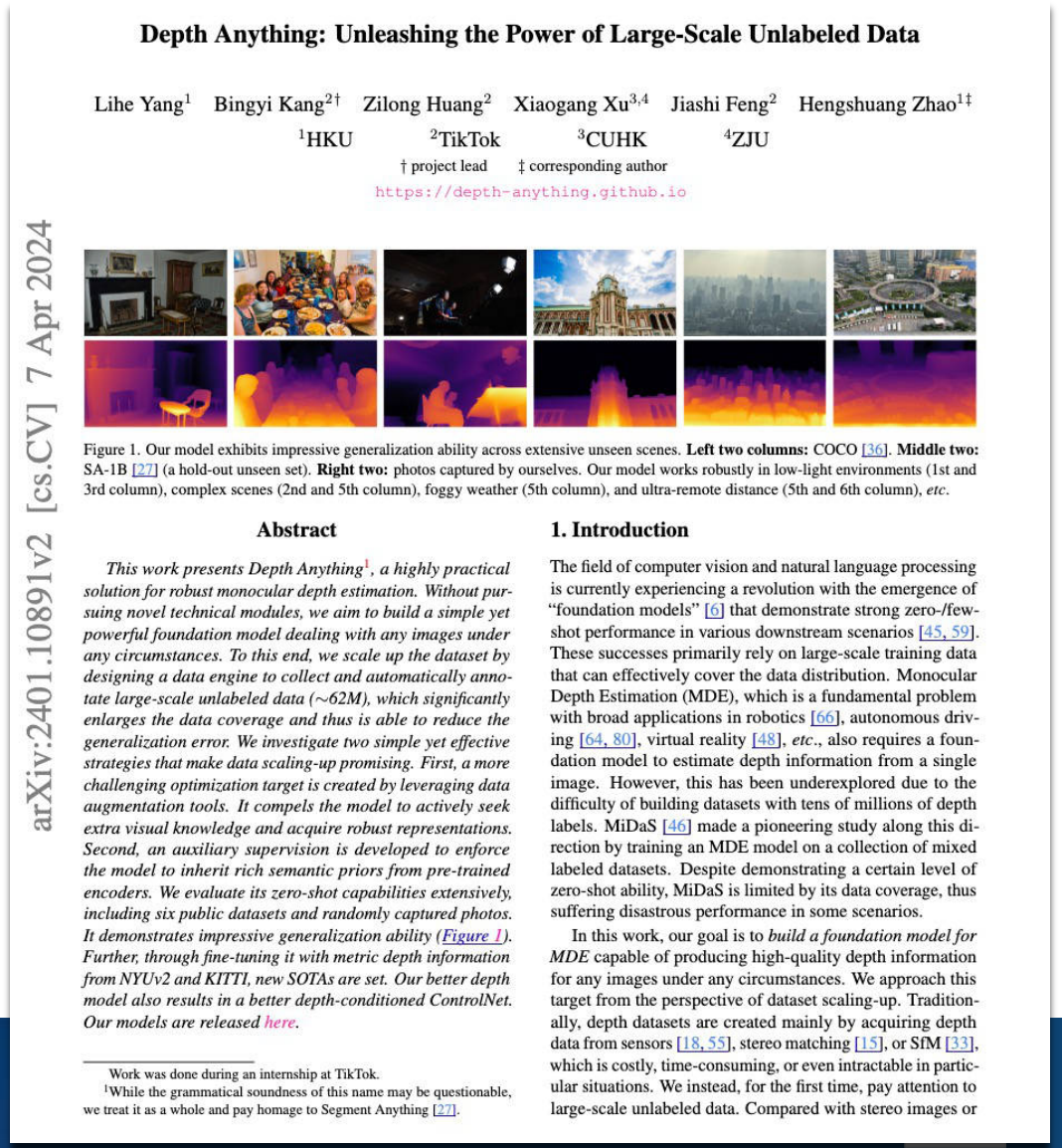
Monocular Depth from Self-Supervision and Pseudo-Labels

CRICOS No.00213J

Monocular Depth from Self-Supervision and Pseudo-Labels

Core Idea:

- Train a depth prediction model in **supervised** way.
  - It is initialised with a DINOv2 encoder, and then fine-tuned.
- Use this model as the "**teacher**".
- The teacher generates "**pseudo-labels**" for the much larger **unlabeled** datasets
  - i.e. we don't have ground truth for them!
- Let the **student** network learn from the pseudo labels



Depth Anything: Unleashing the Power of Large-Scale Unlabeled Data

Lihe Yang[1] Bingyi Kang[2†] Zilong Huang[2] Xiaogang Xu[3,4] Jiashi Feng[2] Hengshuang Zhao[1‡]
[1]HKU [2]TikTok [3]CUHK [4]ZJU
† project lead ‡ corresponding author
https://depth-anything.github.io

arXiv:2401.10891v2 [cs.CV] 7 Apr 2024

Figure 1. Our model exhibits impressive generalization ability across extensive unseen scenes. **Left two columns:** COCO [36]. **Middle two:** SA-1B [27] (a hold-out unseen set). **Right two:** photos captured by ourselves. Our model works robustly in low-light environments (1st and 3rd column), complex scenes (2nd and 5th column), foggy weather (5th column), and ultra-remote distance (5th and 6th column), *etc*.

Abstract

This work presents Depth Anything[1], a highly practical solution for robust monocular depth estimation. Without pursuing novel technical modules, we aim to build a simple yet powerful foundation model dealing with any images under any circumstances. To this end, we scale up the dataset by designing a data engine to collect and automatically annotate large-scale unlabeled data (~62M), which significantly enlarges the data coverage and thus is able to reduce the generalization error. We investigate two simple yet effective strategies that make data scaling-up promising. First, a more challenging optimization target is created by leveraging data augmentation tools. It compels the model to actively seek extra visual knowledge and acquire robust representations. Second, an auxiliary supervision is developed to enforce the model to inherit rich semantic priors from pre-trained encoders. We evaluate its zero-shot capabilities extensively, including six public datasets and randomly captured photos. It demonstrates impressive generalization ability (Figure 1). Further, through fine-tuning it with metric depth information from NYUv2 and KITTI, new SOTAs are set. Our better depth model also results in a better depth-conditioned ControlNet. Our models are released here.

Work was done during an internship at TikTok.
[1]While the grammatical soundness of this name may be questionable, we treat it as a whole and pay homage to Segment Anything [27].

1. Introduction

The field of computer vision and natural language processing is currently experiencing a revolution with the emergence of "foundation models" [6] that demonstrate strong zero-/few-shot performance in various downstream scenarios [45, 59]. These successes primarily rely on large-scale training data that can effectively cover the data distribution. Monocular Depth Estimation (MDE), which is a fundamental problem with broad applications in robotics [66], autonomous driving [64, 80], virtual reality [48], *etc*., also requires a foundation model to estimate depth information from a single image. However, this has been underexplored due to the difficulty of building datasets with tens of millions of depth labels. MiDaS [46] made a pioneering study along this direction by training an MDE model on a collection of mixed labeled datasets. Despite demonstrating a certain level of zero-shot ability, MiDaS is limited by its data coverage, thus suffering disastrous performance in some scenarios.

In this work, our goal is to *build a foundation model for MDE* capable of producing high-quality depth information for any images under any circumstances. We approach this target from the perspective of dataset scaling-up. Traditionally, depth datasets are created mainly by acquiring depth data from sensors [18, 55], stereo matching [15], or SfM [33], which is costly, time-consuming, or even intractable in particular situations. We instead, for the first time, pay attention to large-scale unlabeled data. Compared with stereo images or

Monocular Depth from Self-Supervision and Pseudo-Labels

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Dataset	Indoor	Outdoor	Label	# Images
Labeled Datasets				
BlendedMVS [77]	✓	✓	Stereo	115K
DIML [13]	✓	✓	Stereo	927K
HRWSI [68]	✓	✓	Stereo	20K
IRS [62]	✓		Stereo	103K
MegaDepth [33]		✓	SfM	128K
TartanAir [63]	✓	✓	Stereo	306K
Unlabeled Datasets				
BDD100K [82]		✓	None	8.2M
Google Landmarks [65]		✓	None	4.1M
ImageNet-21K [50]	✓	✓	None	13.1M
LSUN [81]	✓		None	9.8M
Objects365 [53]	✓	✓	None	1.7M
Open Images V7 [30]	✓	✓	None	7.8M
Places365 [88]	✓	✓	None	6.5M
SA-1B [27]	✓	✓	None	11.1M

Table 1. In total, our Depth Anything is trained on 1.5M labeled images and **62M unlabeled images** jointly.

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we failed to gain improvements with such a self-training pipeline …

We conjecture that, with already sufficient labeled images in our case, the extra knowledge acquired from additional unlabeled images is rather limited.

Especially considering the teacher and student share the same pre-training and architecture, they tend to make similar correct or false predictions on the unlabeled set …

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Monocular Depth from Self-Supervision and Pseudo-Labels

To address the dilemma, we propose to challenge the student with a more difficult optimization target for additional visual knowledge on unlabeled images.

We inject strong perturbations to unlabeled images during training. It compels our student model to actively seek extra visual knowledge and acquire invariant representations from these unlabeled images. These advantages help our model deal with the open world more robustly.

We introduce two forms of perturbations: one is strong color distortions, including color jittering and Gaussian blurring, and the other is strong spatial distortion.

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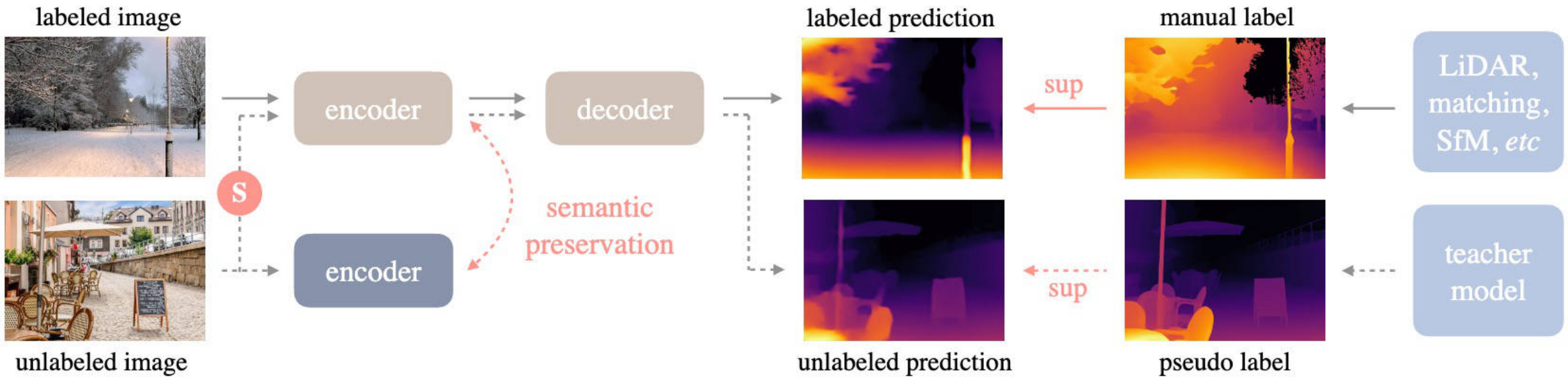


Figure 2. Our pipeline. Solid line: flow of labeled images, dotted line: unlabeled images. We especially highlight the value of large-scale unlabeled images. The S denotes adding strong perturbations (Section 3.2). To equip our depth estimation model with rich semantic priors, we enforce an auxiliary constraint between the online student model and a frozen encoder to preserve the semantic capability (Section 3.3).



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Lihe Yang[1] Bingyi Kang[2†] Zilong Huang[2] Xiaogang Xu[3,4] Jiashi Feng[2] Hengshuang Zhao[1‡]

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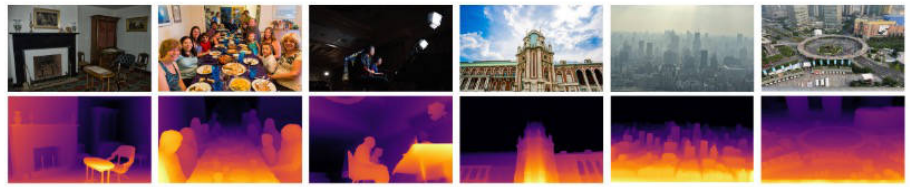


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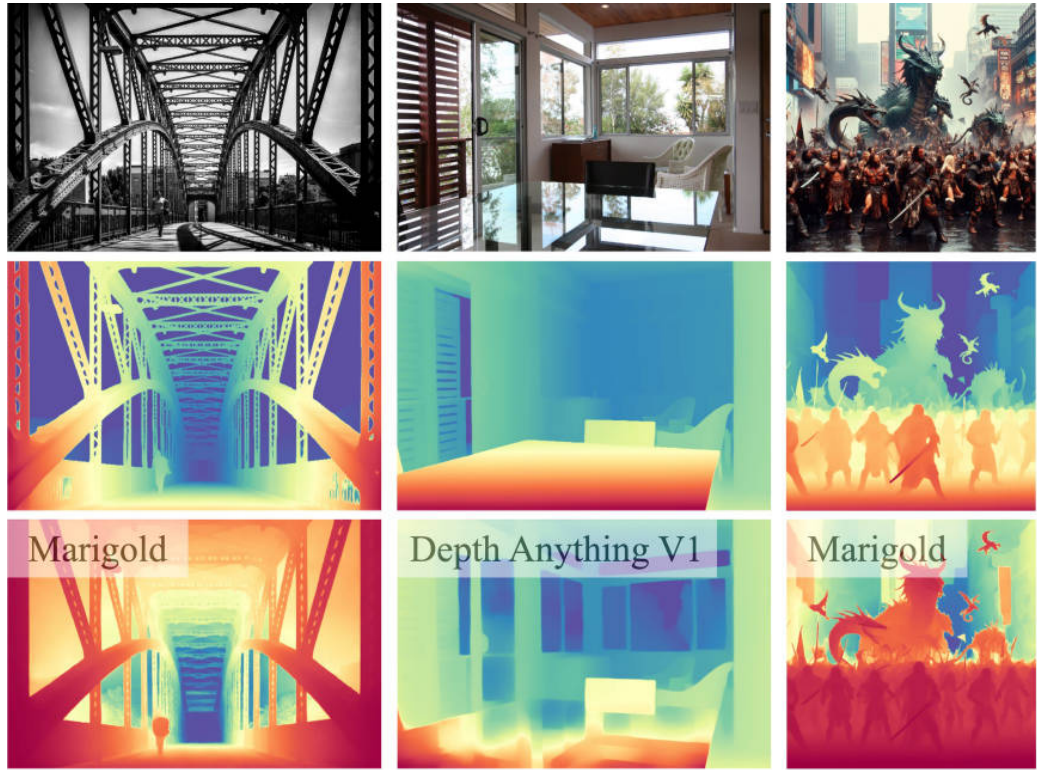
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Latest: DepthAnything V2

- Train the teacher on **synthetic depth** instead of noisy real depth labels. Synthetic data gives extremely clean, dense ground-truth depth and fine geometric detail.
- Use a much larger teacher: DINOv2-Giant. A better teacher produces substantially better pseudo-depth labels.
- Train the final students on 62M+ pseudo-labelled real images. So the teacher learns geometry from clean synthetic data, then transfers that knowledge to the real world through pseudo-labeling



CRICOS No.00213J

Depth Anything V2

Use it yourself!

```python
import requests
from PIL import Image

from transformers import pipeline


# load pipe
pipe = pipeline(task="depth-estimation", model="depth-anything/Depth-Anything-V2-Sr

# load image
url = 'http://images.cocodataset.org/val2017/000000039769.jpg'
image = Image.open(requests.get(url, stream=True).raw)

# inference
depth = pipe(image)["depth"]
```



arXiv:2406.09414v2 [cs.CV] 20 Oct 2024

Depth Anything V2

Lihe Yang[1] Bingyi Kang[2†] Zilong Huang[2]
Zhen Zhao Xiaogang Xu Jiashi Feng[2] Hengshuang Zhao[1‡]
[1]HKU [2]TikTok
[†]project lead [‡]corresponding author
https://depth-anything-v2.github.io

Figure 1: Depth Anything V2 significantly outperforms V1 [89] in robustness and fine-grained details. Compared with SD-based models [31, 25], it enjoys faster inference speed, fewer parameters, and higher depth accuracy.

Abstract

This work presents *Depth Anything V2*. Without pursuing fancy techniques, we aim to reveal crucial findings to pave the way towards building a powerful monocular depth estimation model. Notably, compared with V1 [89], this version produces much finer and more robust depth predictions through three key practices: 1) replacing all labeled real images with synthetic images, 2) scaling up the capacity of our teacher model, and 3) teaching student models via the bridge of large-scale pseudo-labeled real images. Compared with the latest models [31] built on Stable Diffusion, our models are significantly more efficient (more than 10× faster) and more accurate. We offer models of different scales (ranging from 25M to 1.3B params) to support extensive scenarios. Benefiting from their strong generalization capability, we fine-tune them with metric depth labels to obtain our metric depth models. In addition to our models, considering the limited diversity and frequent noise in current test sets, we construct a versatile evaluation benchmark with precise annotations and diverse scenes to facilitate future research.

Work done during an internship at TikTok.

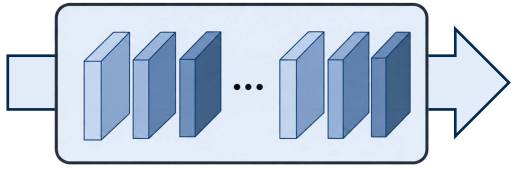
Or try the demo: https://huggingface.co/spaces/depth-anything/Depth-Anything-V2

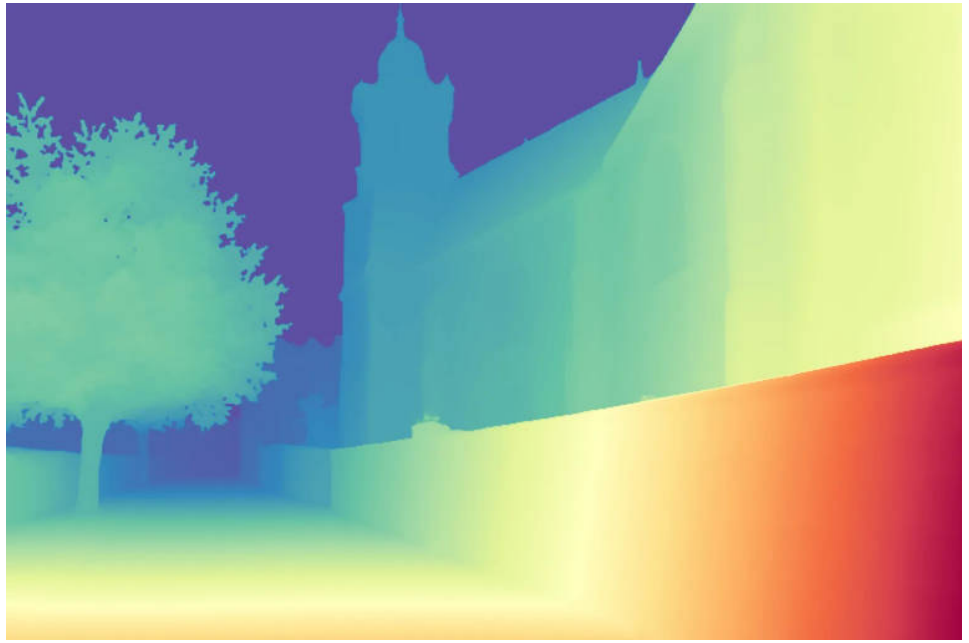
From Depth to Learned Multi-View Geometry

Depth Map:
The Z-Coordinate for Every Pixel

Depth Map: for each pixel, output the Z-coordinate of the corresponding 3D point in the camera coordinate system.







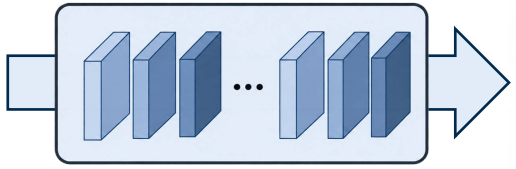
H x W x 1

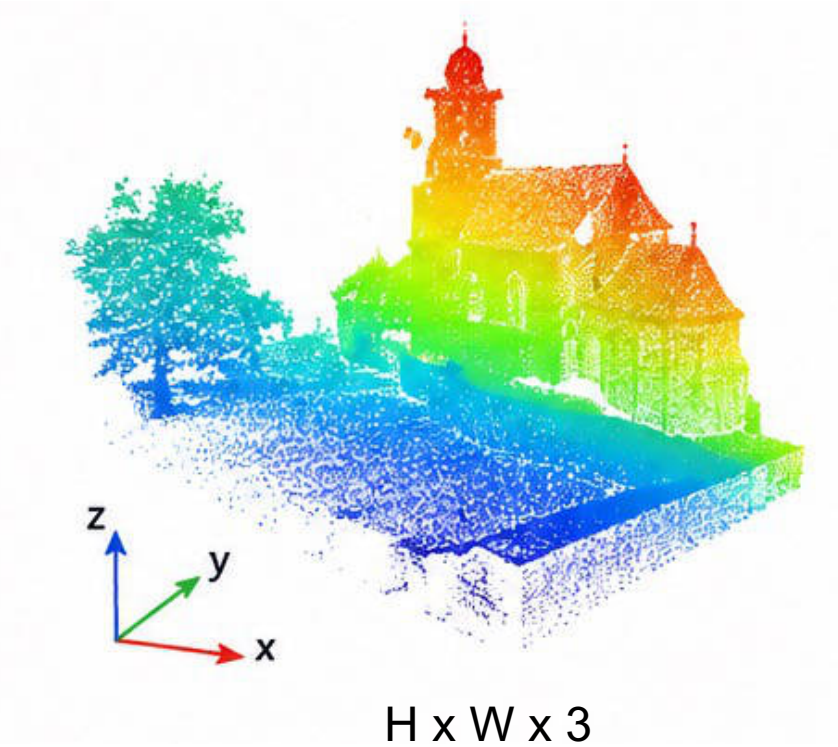
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Point Map: A 3D Point for Every Pixel

Point Map:
for each pixel output the 3D coordinate (x,y,z)







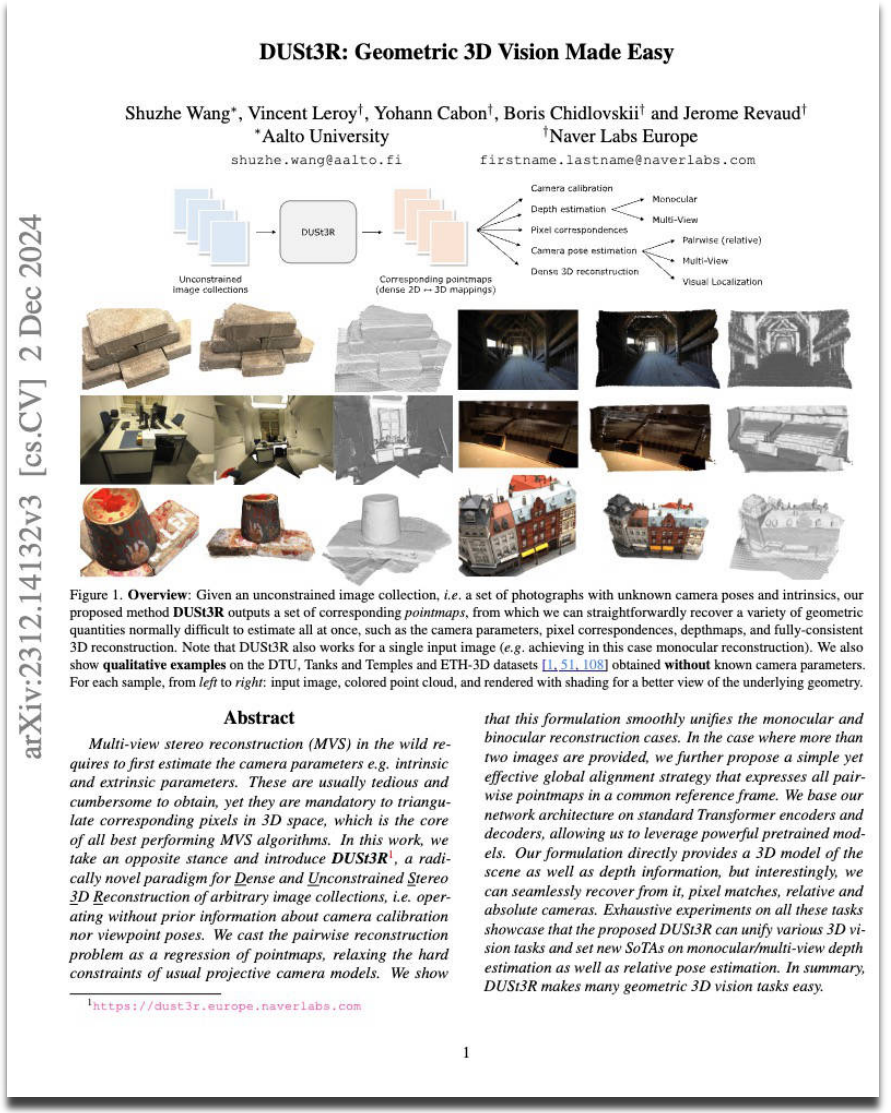
H x W x 3

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DUSt3R

*"Given a **set of photographs with unknown camera poses and intrinsics**, DUSt3R **outputs a set of corresponding pointmaps**, from which we can straightforwardly recover a variety of geometric quantities normally difficult to estimate all at once, such as the camera parameters, pixel correspondences, depthmaps, and fully-consistent 3D reconstruction.*

Note that DUSt3R also works for a single input image (e.g. achieving in this case monocular reconstruction)."



DUSt3R: Geometric 3D Vision Made Easy

Shuzhe Wang*, Vincent Leroy†, Yohann Cabon†, Boris Chidlovskii† and Jerome Revaud†
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shuzhe.wang@aalto.fi firstname.lastname@naverlabs.com

arXiv:2312.14132v3 [cs.CV] 2 Dec 2024

Figure 1. **Overview**: Given an unconstrained image collection, *i.e.* a set of photographs with unknown camera poses and intrinsics, our proposed method **DUSt3R** outputs a set of corresponding *pointmaps*, from which we can straightforwardly recover a variety of geometric quantities normally difficult to estimate all at once, such as the camera parameters, pixel correspondences, depthmaps, and fully-consistent 3D reconstruction. Note that DUSt3R also works for a single input image (*e.g.* achieving in this case monocular reconstruction). We also show **qualitative examples** on the DTU, Tanks and Temples and ETH-3D datasets [1, 51, 108] obtained **without** known camera parameters. For each sample, from *left* to *right*: input image, colored point cloud, and rendered with shading for a better view of the underlying geometry.

Abstract

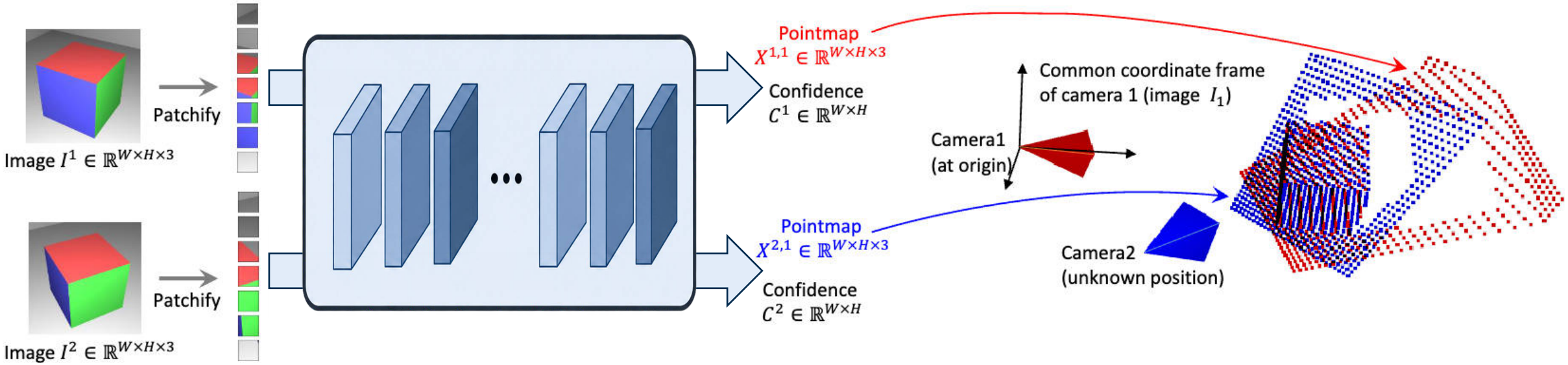
*Multi-view stereo reconstruction (MVS) in the wild requires to first estimate the camera parameters e.g. intrinsic and extrinsic parameters. These are usually tedious and cumbersome to obtain, yet they are mandatory to triangulate corresponding pixels in 3D space, which is the core of all best performing MVS algorithms. In this work, we take an opposite stance and introduce **DUSt3R**[1], a radically novel paradigm for Dense and Unconstrained Stereo 3D Reconstruction of arbitrary image collections, i.e. operating without prior information about camera calibration nor viewpoint poses. We cast the pairwise reconstruction problem as a regression of pointmaps, relaxing the hard constraints of usual projective camera models. We show that this formulation smoothly unifies the monocular and binocular reconstruction cases. In the case where more than two images are provided, we further propose a simple yet effective global alignment strategy that expresses all pairwise pointmaps in a common reference frame. We base our network architecture on standard Transformer encoders and decoders, allowing us to leverage powerful pretrained models. Our formulation directly provides a 3D model of the scene as well as depth information, but interestingly, we can seamlessly recover from it, pixel matches, relative and absolute cameras. Exhaustive experiments on all these tasks showcase that the proposed DUSt3R can unify various 3D vision tasks and set new SoTAs on monocular/multi-view depth estimation as well as relative pose estimation. In summary, DUSt3R makes many geometric 3D vision tasks easy.*

[1] https://dust3r.europe.naverlabs.com





DUSt3R



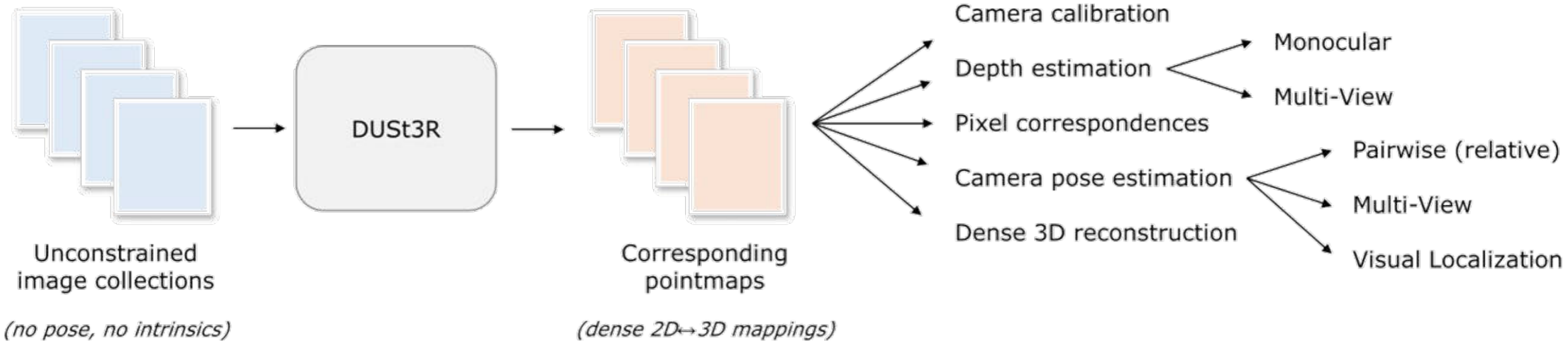
The predicted point maps are consistent, but not necessarily metrically correct, i.e. up-to-scale.

DUSt3R

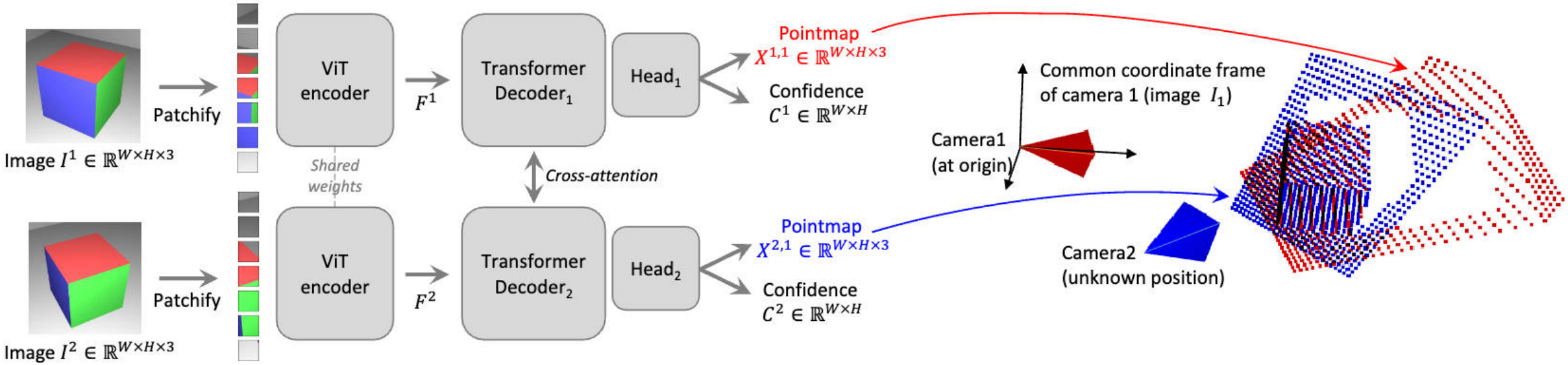
DUSt3R's key conceptual shift is that it does **not start by explicitly finding 2D feature matches, estimating the essential matrix, triangulating, etc.**

Instead, given two images, it predicts **dense 3D pointmaps**.

From those pointmaps, you can subsequently recover depth, correspondences, relative camera pose, focal length, and so on.



DUSt3R



CRICOS No.00213J

DUSt3R:
Geometric 3D Vision Made Easy

S. Wang [1], V. Leroy [2], Y. Cabon [2], B. Chidlovskii [2] and J. Revaud [2]

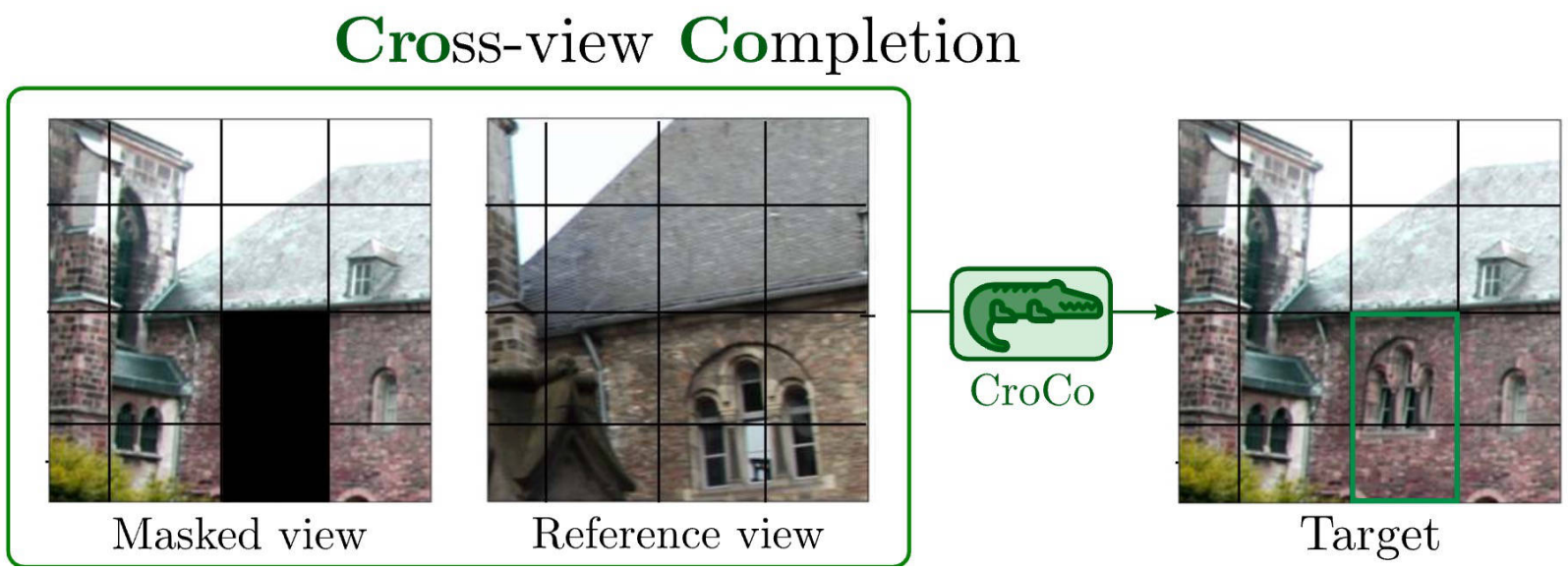
[1] Aalto University

2 Naver Labs Europe

DUSt3R

How is it trained?

- Initialised with a network trained on a different task: **Cross-View Completion**



CRICOS No.00213J

CroCo: Self-Supervised Pre-training for 3D Vision Tasks by Cross-View Completion

Philippe Weinzaepfel Vincent Leroy Thomas Lucas
Romain Brégier Yohann Cabon Vaibhav Arora Leonid Antsfeld
Boris Chidlovskii Gabriela Csurka Jérôme Revaud
NAVER LABS Europe
https://europe.naverlabs.com/research/computer-vision/croco/

Abstract

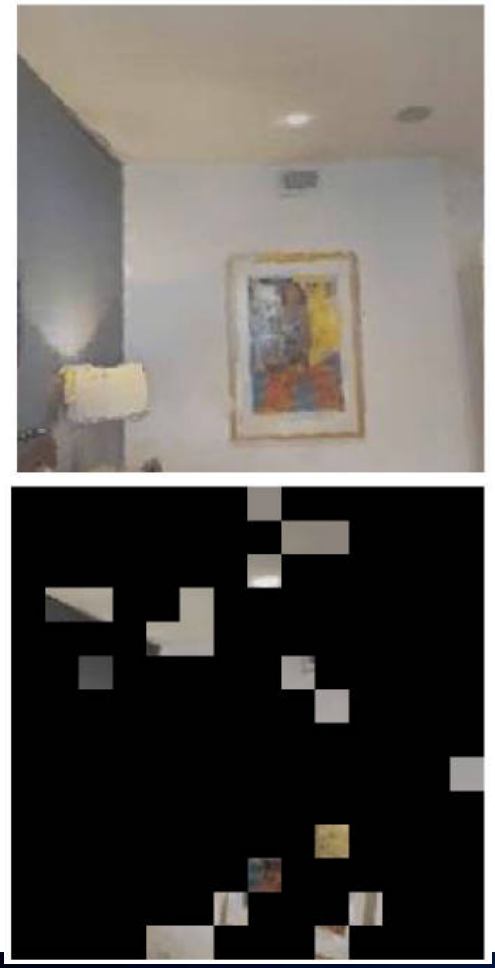
Masked Image Modeling (MIM) has recently been established as a potent pre-training paradigm. A pretext task is constructed by masking patches in an input image, and this masked content is then predicted by a neural network using visible patches as sole input. This pre-training leads to state-of-the-art performance when finetuned for high-level semantic tasks, *e.g.* image classification and object detection. In this paper we instead seek to learn representations that transfer well to a wide variety of 3D vision and lower-level geometric downstream tasks, such as depth prediction or optical flow estimation. Inspired by MIM, we propose an unsupervised representation learning task trained from *pairs* of images showing the same scene from different viewpoints. More precisely, we propose the pretext task of *cross-view completion* where the first input image is partially masked, and this masked content has to be reconstructed from the visible content and the second image. In single-view MIM, the masked content often cannot be inferred precisely from the visible portion only, so the model learns to act as a prior influenced by high-level semantics. In contrast, this ambiguity can be resolved with cross-view completion from the second unmasked image, on the condition that the model is able to understand the spatial relationship between the two images. Our experiments show that our pretext task leads to significantly improved performance for monocular 3D vision downstream tasks such as depth estimation. In addition, our model can be directly applied to binocular downstream tasks like optical flow or relative camera pose estimation, for which we obtain competitive results without bells and whistles, *i.e.*, using a generic architecture without any task-specific design.

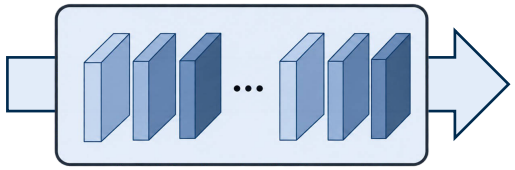


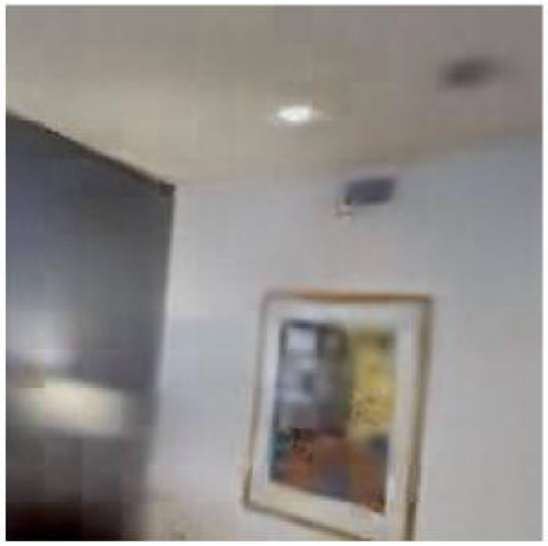
Figure 1: **Auto-completion *vs.* Cross-view completion tasks.** *Left:* given a masked image, a model trained for auto-completion can only leverage the visible context to fill-in the blanks and thus relies mainly on high-level semantic information. *Right:* given an additional view of the same scene, cross-view completion makes precise reconstruction possible, assuming that the model is able to understand both the scene geometry and the spatial relationship between the two images.

36th Conference on Neural Information Processing Systems (NeurIPS 2022).

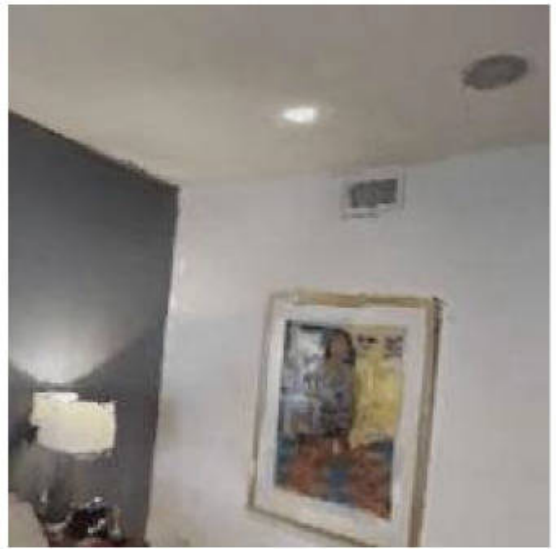
Cross-View Completion







Prediction



Ground Truth

DUSt3R

How is it trained?

- Initialised with a network trained on a different task: **Cross-View Completion**
- Then train using direct 3D supervision on image pairs with GT point maps
- Use datasets that allow constructing the ground-truth 3D point for each pixel
  - e.g. from depth, synthetic scene geometry, known camera poses

Datasets	Type	N Pairs
Habitat [104]	Indoor / Synthetic	1000k
CO3Dv2 [94]	Object-centric	941k
ScanNet++ [166]	Indoor / Real	224k
ArkitScenes [25]	Indoor / Real	2040k
Static Thing 3D [110]	Object / Synthetic	337k
MegaDepth [56]	Outdoor / Real	1761k
BlendedMVS [162]	Outdoor / Synthetic	1062k
Waymo [122]	Outdoor / Real	1100k

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DUSt3R

How does it align more than 2 cameras?

- Calculate pairwise alignment
- Then perform global alignment via a big optimisation problem

Global optimization. We use the connectivity graph $\mathcal{G}$ to recover *globally aligned* pointmaps $\{\chi^n \in \mathbb{R}^{W\times H\times 3}\}$ for all cameras $n = 1\dots N$. To that aim, we first predict, for each image pair $e = (n, m) \in \mathcal{E}$, the pairwise pointmaps $X^{n,n}, X^{m,n}$ and their associated confidence maps $C^{n,n}, C^{m,n}$. For the sake of clarity, let us define $X^{n,e} := X^{n,n}$ and $X^{m,e} := X^{m,n}$. Since our goal involves to express all pairwise predictions in a common coordinate frame, we introduce a pairwise pose $P_e \in \mathbb{R}^{3\times 4}$ and scaling $\sigma_e > 0$ associated to each pair $e \in \mathcal{E}$. We then formulate the following optimization problem:

$$\chi^* = \underset{\chi,P,\sigma}{\arg\min} \sum_{e\in\mathcal{E}} \sum_{v\in e} \sum_{i=1}^{HW} C_i^{v,e} \left\| \chi_i^v - \sigma_e P_e X_i^{v,e} \right\| . \quad (5)$$

Here, with some abuse of notation, we write $v \in e$ for $v \in \{n, m\}$ if $e = (n, m)$. The idea is that, for a given pair e, the *same* rigid transformation P_e should align both pointmaps

MAST3R = DUST3R + Matching



Grounding Image Matching in 3D with MASt3R

Vincent Leroy, Yohann Cabon, and Jerome Revaud

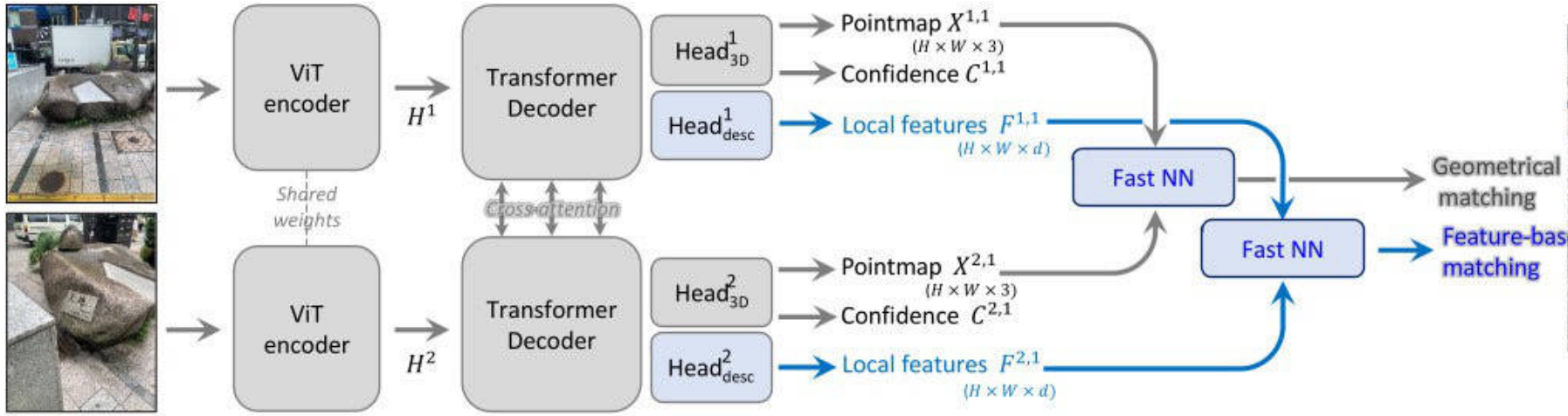
Naver Labs Europe
<firstname>.<lastname>@naverlabs.com

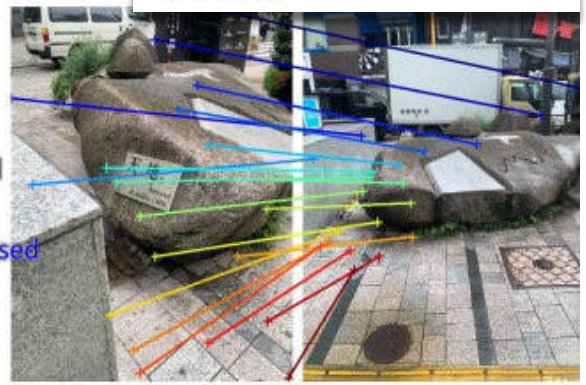
Abstract. Image Matching is a core component of all best-performing algorithms and pipelines in 3D vision. Yet despite matching being fundamentally a 3D problem, intrinsically linked to camera pose and scene geometry, it is typically treated as a 2D problem. This makes sense as the goal of matching is to establish correspondences between 2D pixel fields, but also seems like a potentially hazardous choice. In this work, we take a different stance and propose to cast matching as a 3D task with DUSt3R, a recent and powerful 3D reconstruction framework based on Transformers. Based on pointmaps regression, this method displayed impressive robustness in matching views with extreme viewpoint changes, yet with limited accuracy. We aim here to improve the matching capabilities of such an approach while preserving its robustness. We thus propose to augment the DUSt3R network with a new head that outputs dense local features, trained with an additional matching loss. We further address the issue of quadratic complexity of dense matching, which becomes prohibitively slow for downstream applications if not treated carefully. We introduce a fast reciprocal matching scheme that not only accelerates matching by orders of magnitude, but also comes with theoretical guarantees and, lastly, yields improved results. Extensive experiments show that our approach, coined MASt3R, significantly outperforms the state of the art on multiple matching tasks. In particular, it largely outperforms the best published methods on the challenging Map-free localization dataset.

1 Introduction

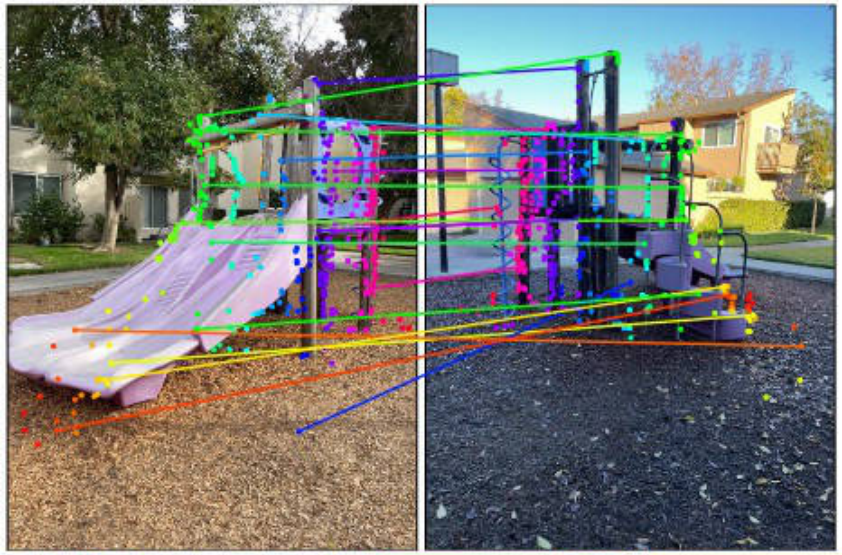
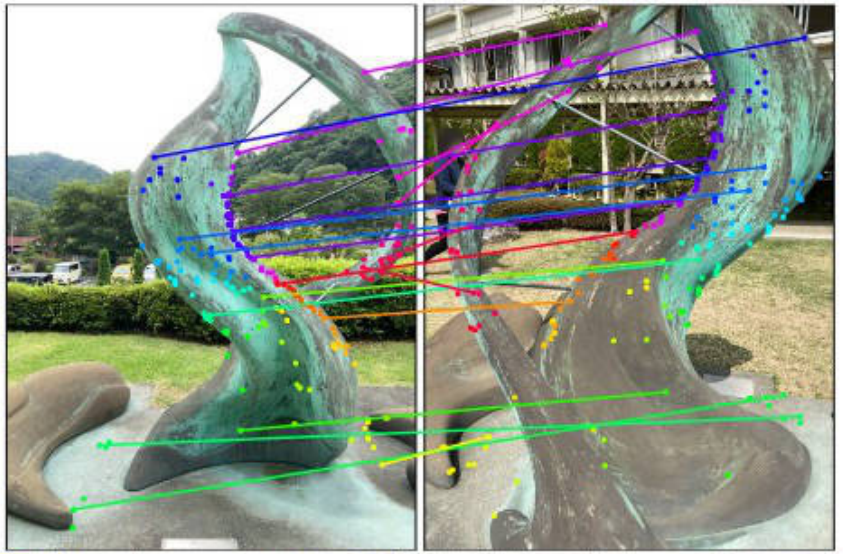
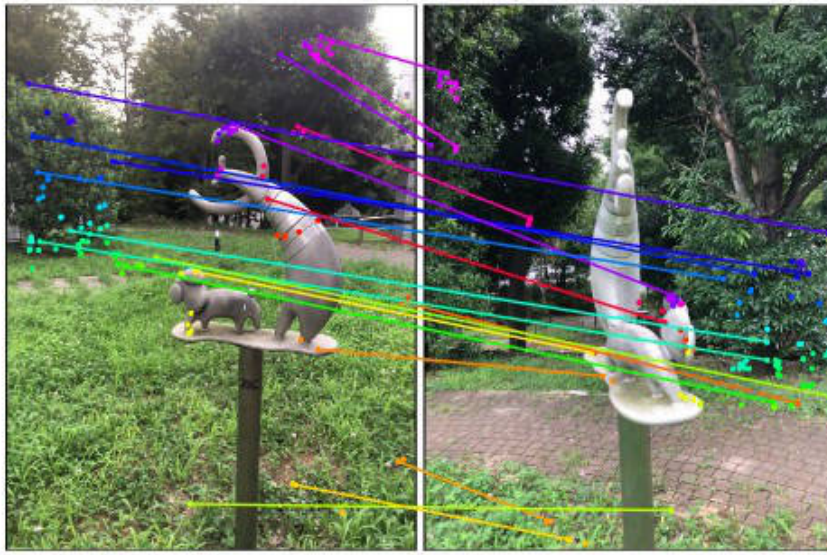
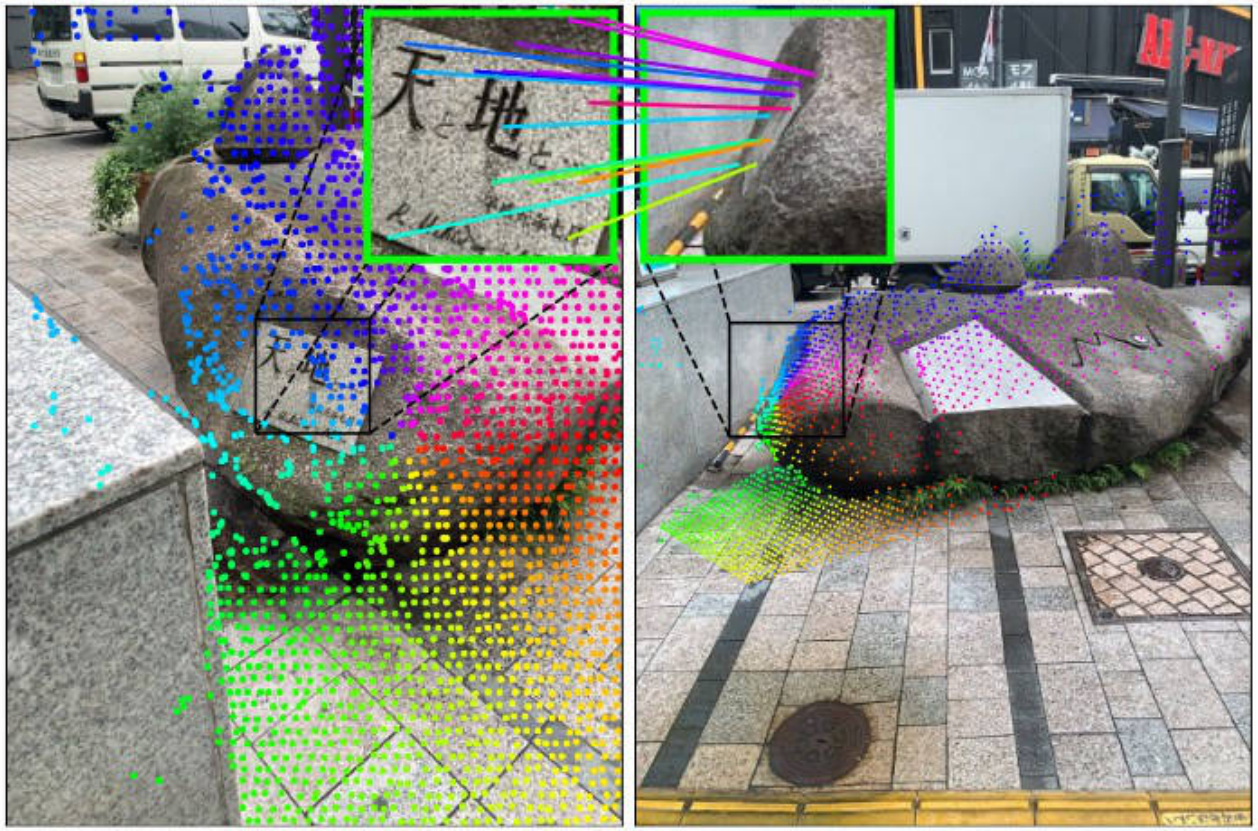
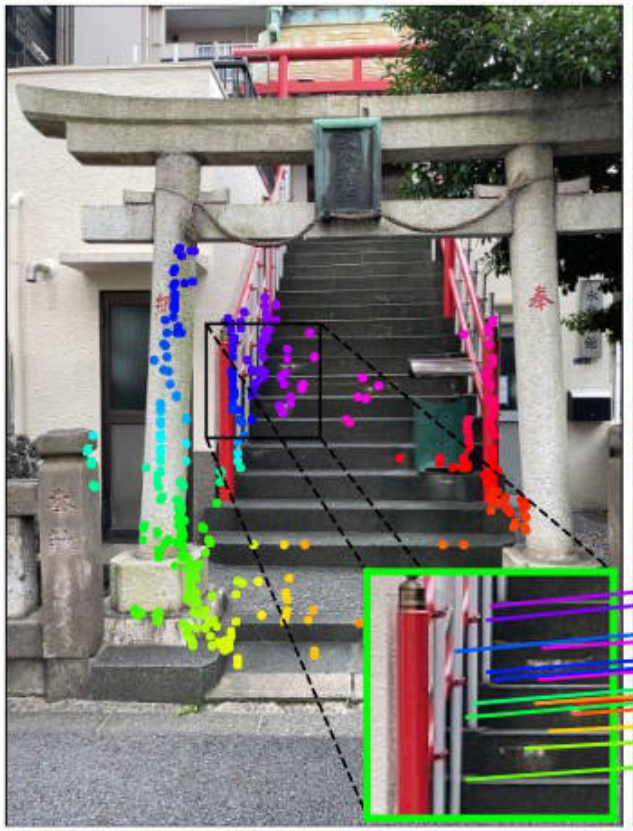
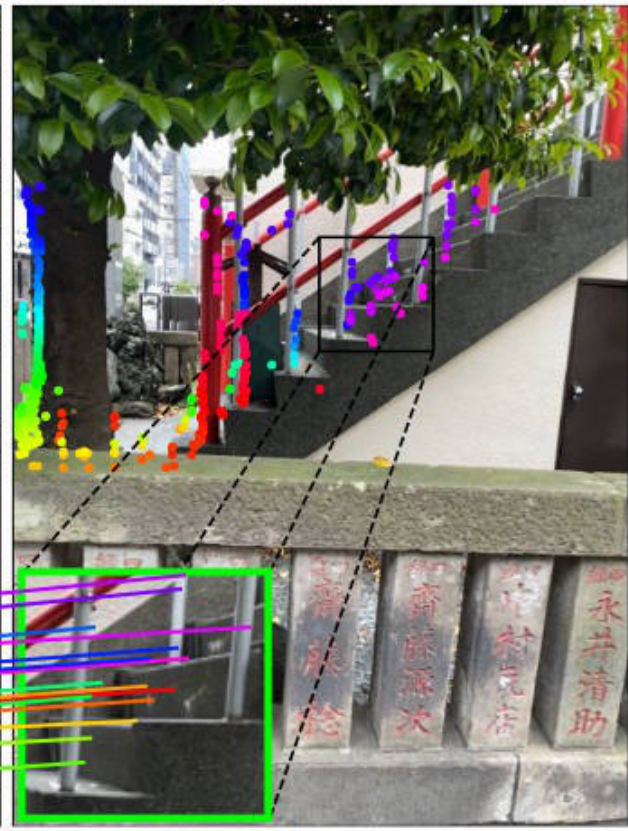
Being able to establish correspondences between pixels across different images of the same scene, denoted as *image matching*, constitutes a core component of all 3D vision applications, spanning mapping [15, 63], localization [43, 78], navigation [16], photogrammetry [35, 68] and autonomous robotics in general [67, 92]. State-of-the-art methods for visual localization, for instance, overwhelmingly rely upon image matching during the offline mapping stage, e.g. using COLMAP [61], as well as during the online localization step, typically using PnP [21]. In this paper, we focus on this core task and aim at producing, given two images, a list of pairwise correspondences, denoted as *matches*. In particular, we seek to output highly accurate and dense matches that are robust to viewpoint and illumination changes because these are, in the end, the limiting factor for real-world applications [18].

Pointmaps are now metric, loss takes true scale of the scene into account!





New: per-pixel descriptor

Map-Free Relocalization

Task: Metric camera localization from an image pair

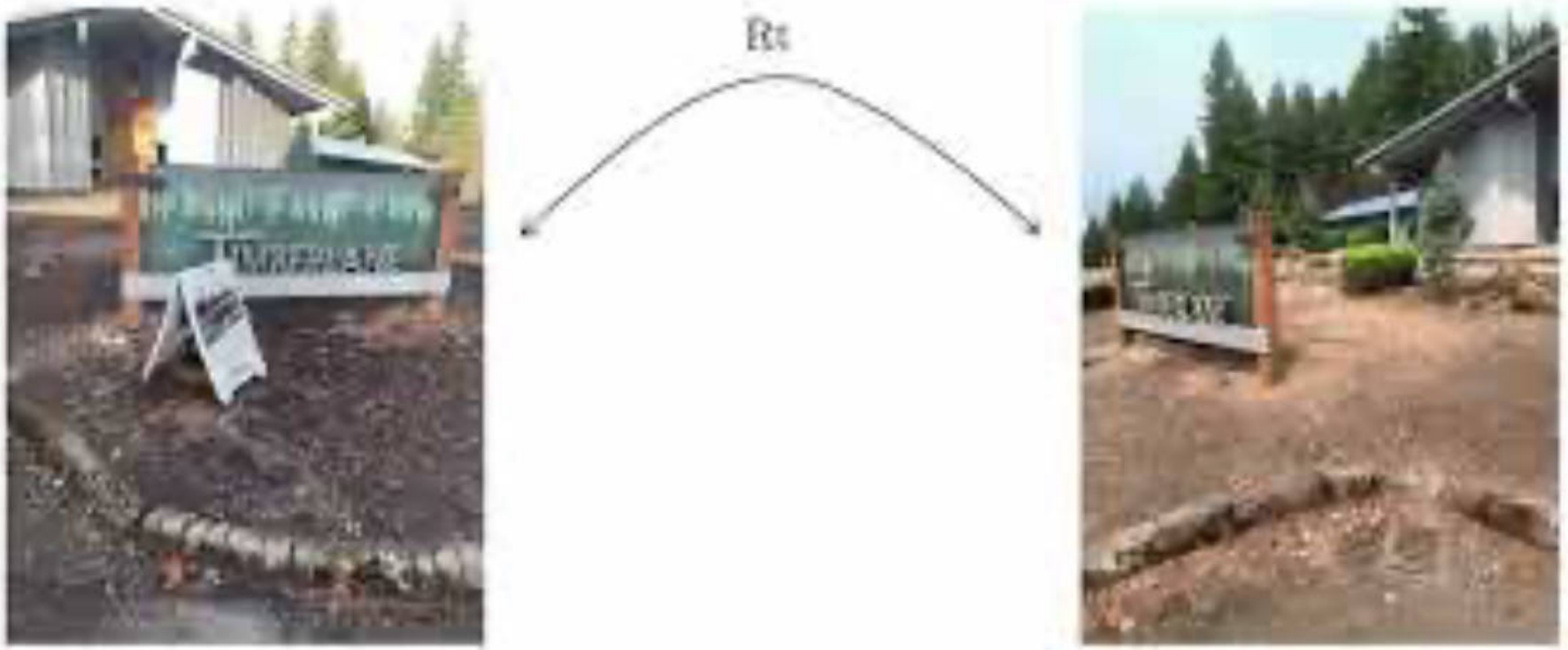


Image 1

Image 2

Translation is metric → Pixel matching alone cannot do this
Use DPT-KITTI for absolute monocular depth prediction

Summary

- **Monocular depth** predicts scene geometry from a single image, but scale and consistency across views are challenging.
- **DUSt3R** predicts a dense 3D pointmap for every pixel from an image pair.
- **MASt3R** builds on DUSt3R by adding dense learned descriptors, improving precise matching and correspondence.

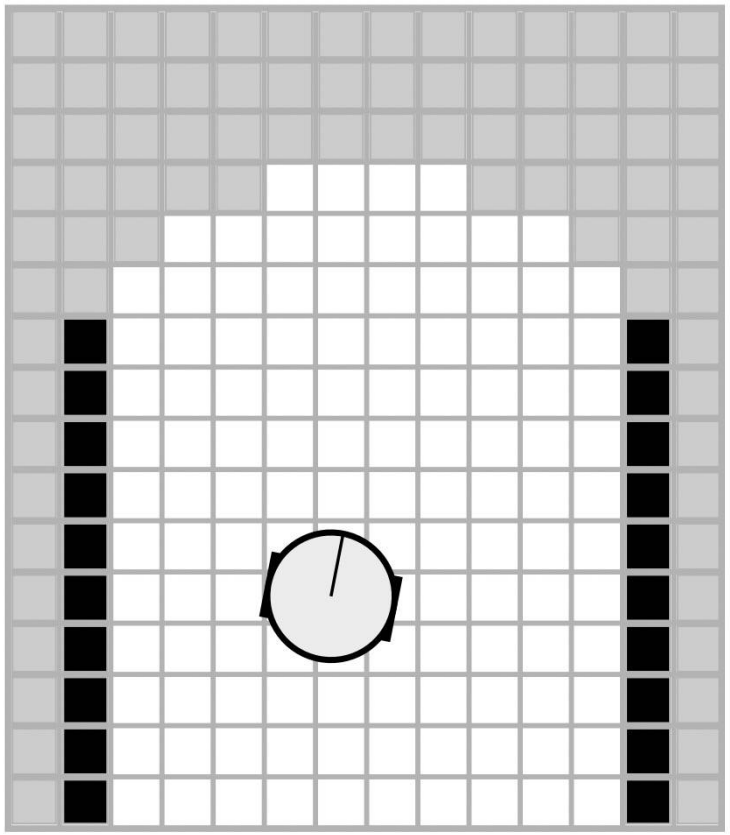
- Learned multi-view methods can replace parts of the classical pipeline:
  features → matches → pose → triangulation

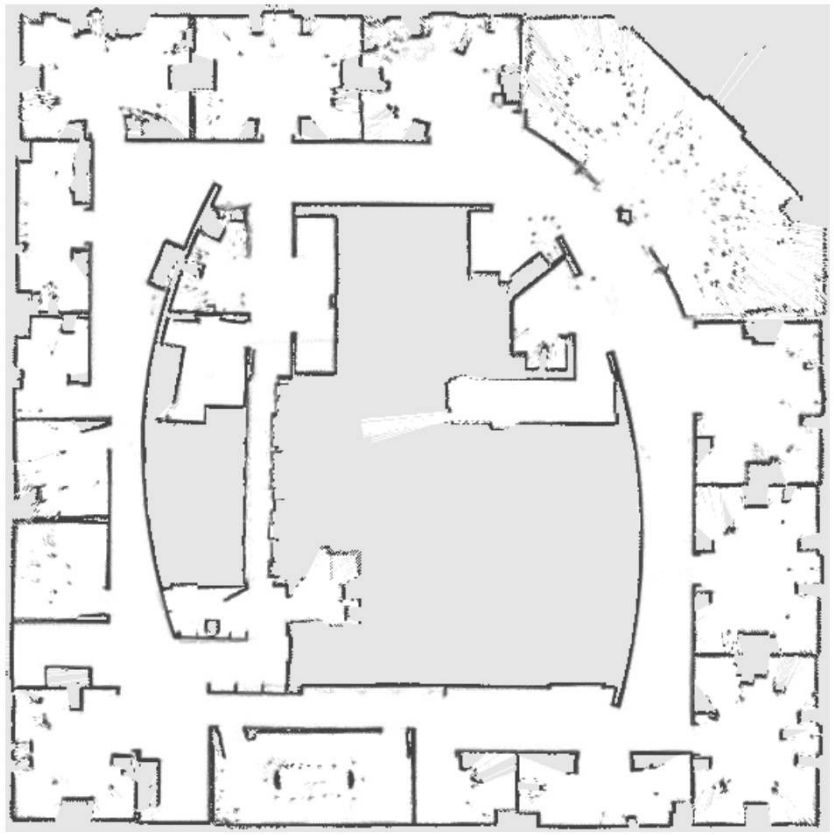
Next Question:
Once we have recovered 3D geometry, how should we represent the scene?

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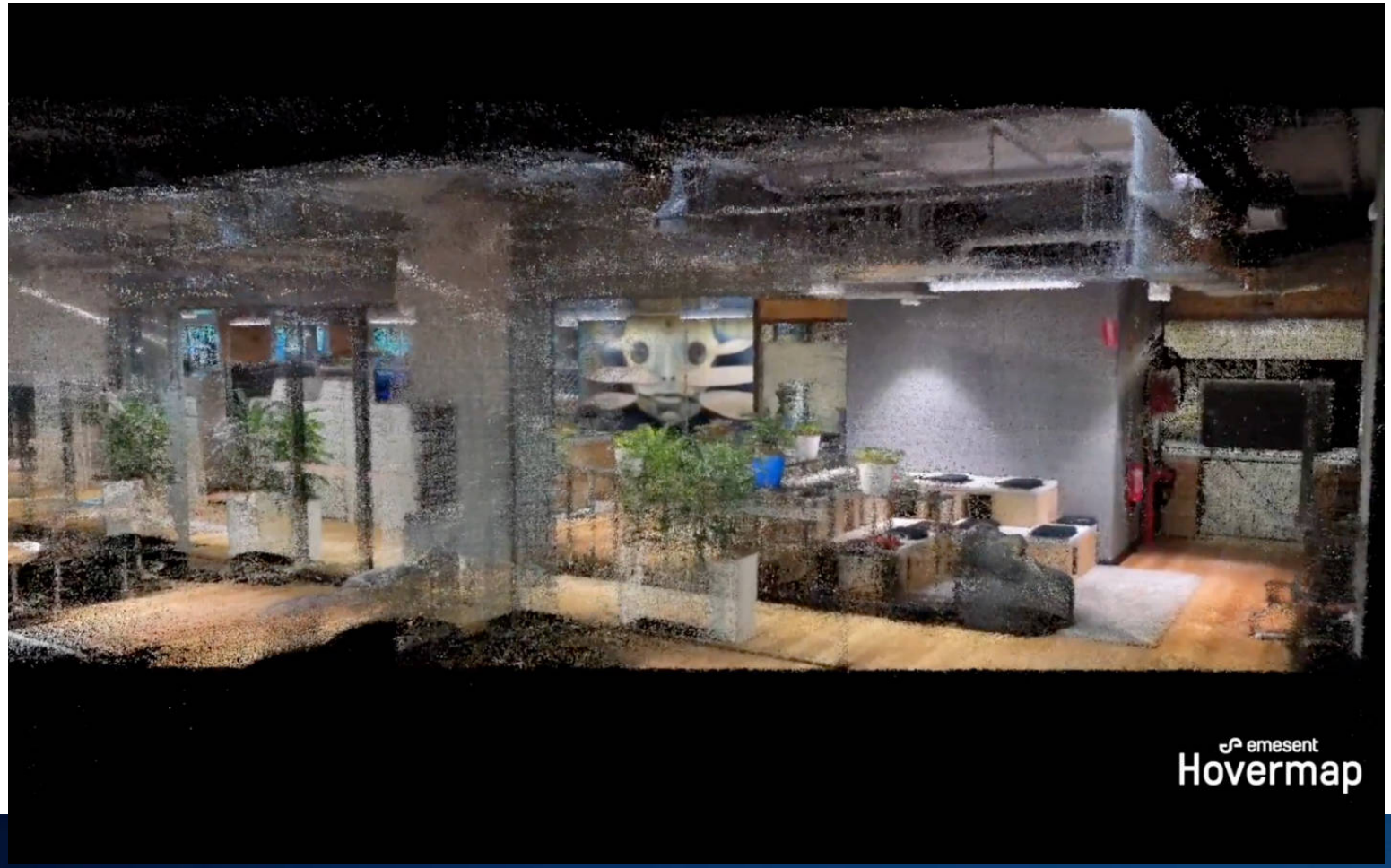
Reminder:
Classical 3D Representations

Occupancy Gridmaps

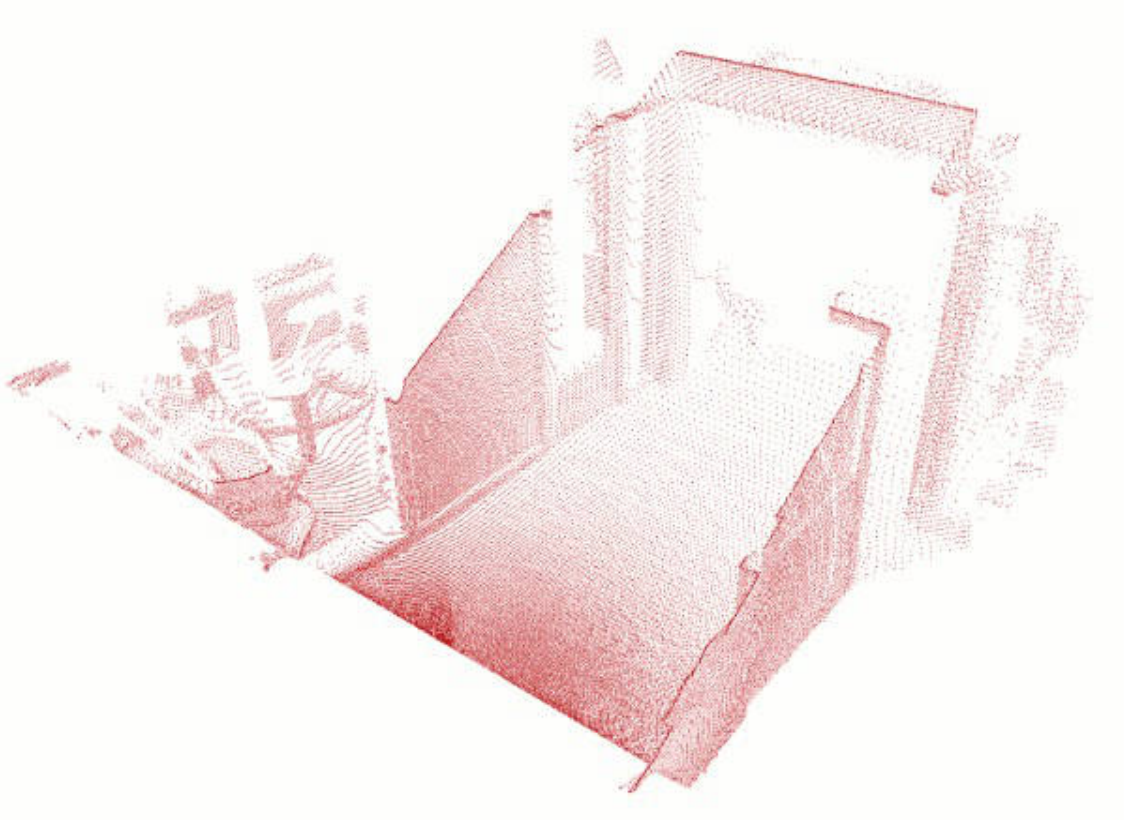


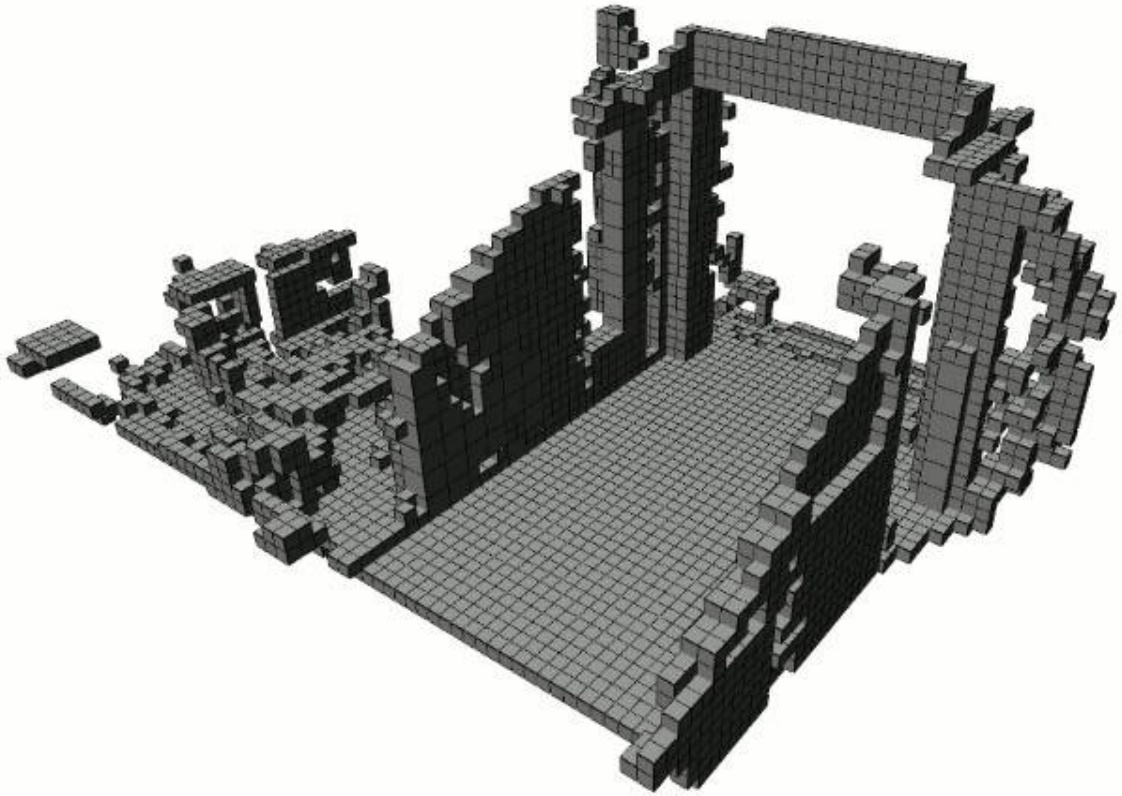


Point Clouds

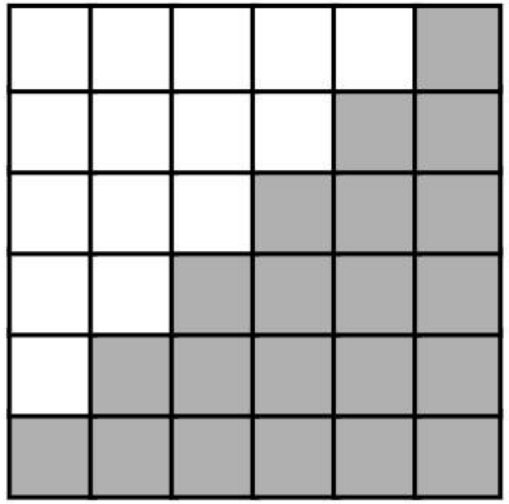


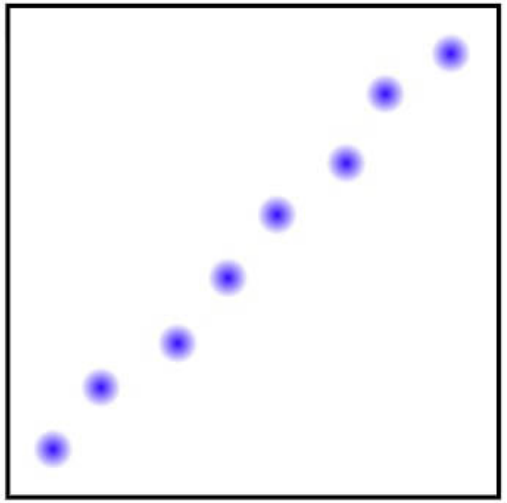
Voxels

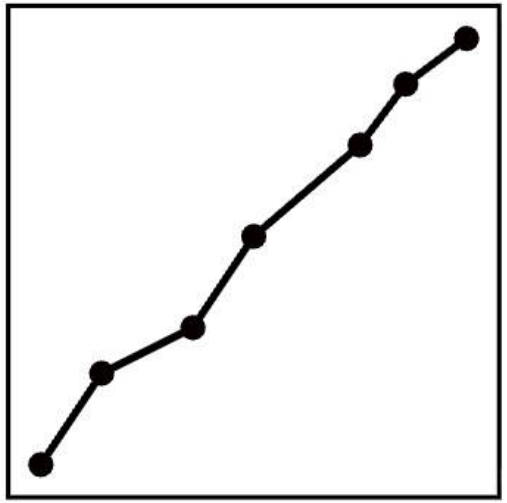




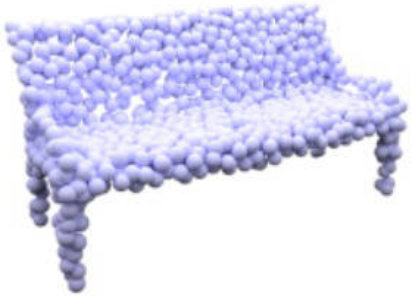
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(a) Voxel (b) Point (c) Mesh

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Octree

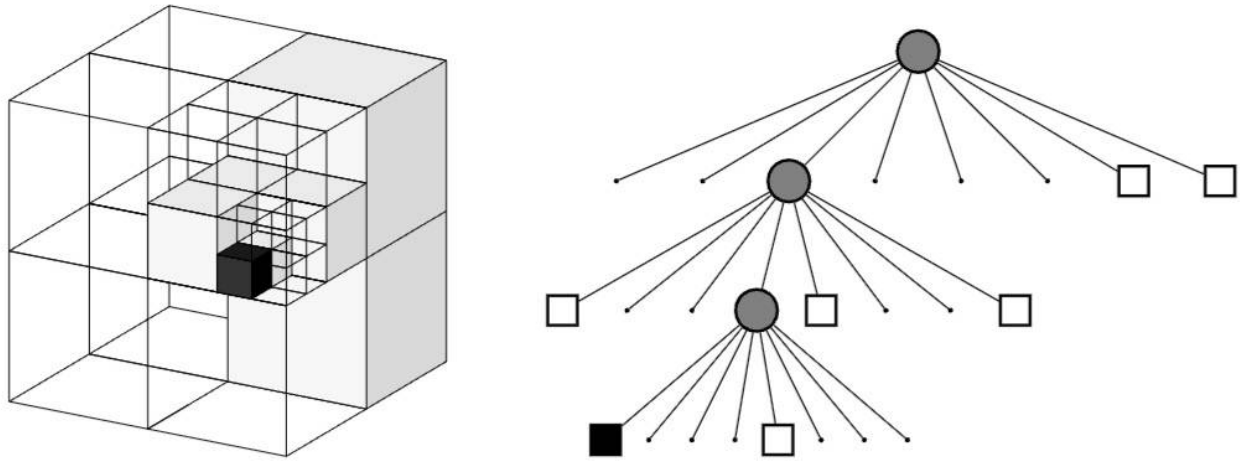


Fig. 2 Example of an octree storing free (shaded white) and occupied (black) cells. The volumetric model is shown on the left and the corresponding tree representation on the right.

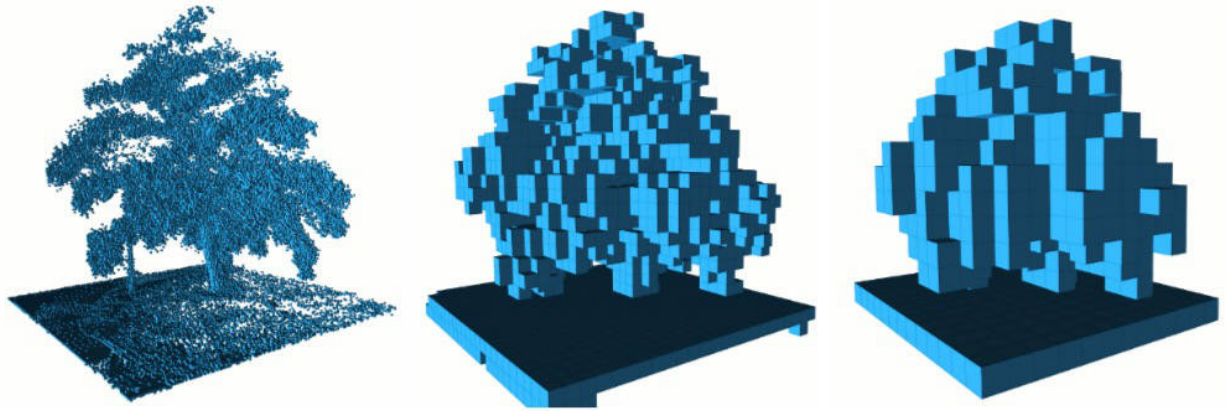


Fig. 3 By limiting the depth of a query, multiple resolutions of the same map can be obtained at any time. Occupied voxels are displayed in resolutions 0.08 m, 0.64 , and 1.28 m.

OctoMap:
An Efficient Probabilistic 3D Mapping Framework Based on Octrees

Armin Hornung · Kai M. Wurm · Maren Bennewitz · Cyrill Stachniss · Wolfram Burgard

Received: 30 April 2012 / Accepted: 31 December 2012

Abstract Three-dimensional models provide a volumetric representation of space which is important for a variety of robotic applications including flying robots and robots that are equipped with manipulators. In this paper, we present an open-source framework to generate volumetric 3D environment models. Our mapping approach is based on octrees and uses probabilistic occupancy estimation. It explicitly represents not only occupied space, but also free and unknown areas. Furthermore, we propose an octree map compression method that keeps the 3D models compact. Our framework is available as an open-source C++ library and has already been successfully applied in several robotics projects. We present a series of experimental results carried out with real robots and on publicly available real-world datasets. The results demonstrate that our approach is able to update the representation efficiently and models the data consistently while keeping the memory requirement at a minimum.

Keywords 3D · Probabilistic · Mapping · Navigation

1 Introduction

Several robotic applications require a 3D model of the environment. These include airborne, underwater, outdoor, or extra-terrestrial missions. However, 3D models are also relevant for many domestic scenarios, for example, for mobile manipulation and also navigation tasks.

This work has been supported by the German Research Foundation (DFG) under contract number SFB/TR-8 and by the European Commission under grant agreement numbers FP7-248258-First-MM and FP7-600890-ROVINA.

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Although 3D mapping is an integral component of many robotic systems, there exist few readily available, reliable, and efficient implementations. The lack of such implementations leads to the re-creation of basic software components and, thus, can be seen as a bottleneck in robotics research. We therefore believe that the development of an open-source 3D mapping framework will greatly facilitate the development of robotic systems that require a three-dimensional geometric representation of the environment.

Most robotics applications require a probabilistic representation, modeling of free, occupied, and unmapped areas, and additionally efficiency with respect to runtime and memory usage. We will now discuss these three requirements in detail.

- **Probabilistic representation:** To create 3D maps, mobile robots sense the environment by taking 3D range measurements. Such measurements are afflicted with uncertainty: Typically, the error in the range measurements is in the order of centimeters. But there may also be seemingly random measurements that are caused by reflections or dynamic obstacles. When the task is to create an accurate model of the environment from such noisy measurements, the underlying uncertainty has to be taken into account probabilistically. Multiple uncertain measurements can then be fused into a robust estimate of the true state of the environment. Another important aspect is that probabilistic sensor fusion allows for the integration of data from multiple sensors and of multiple robots.
- **Modeling of unmapped areas:** In autonomous navigation tasks, a robot can plan collision-free paths only for those areas that have been covered by sensor measurements and detected to be free. Unmapped areas, in contrast, need to be avoided and for this reason the map has to represent such areas. Furthermore, the knowledge

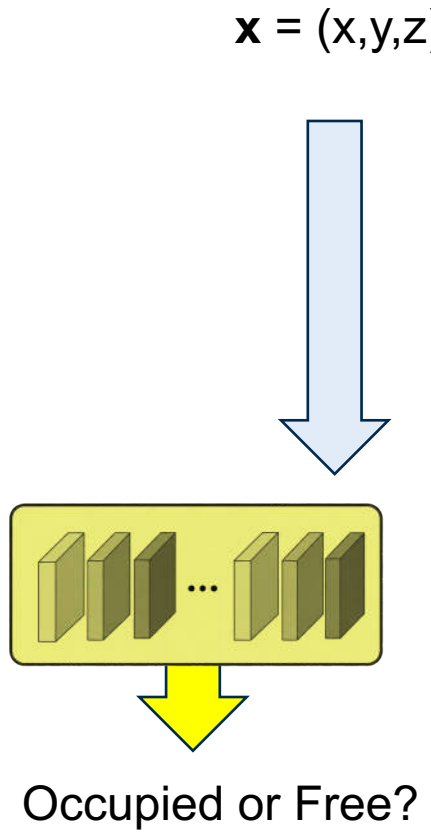
Learned Implicit Geometry

Can the 3D Representation be Learned?

Occupancy Networks

Idea:

- Instead of storing a voxel grid, or a point cloud …
- … can we train a network to predict if a point (x,y,z) is occupied or not?

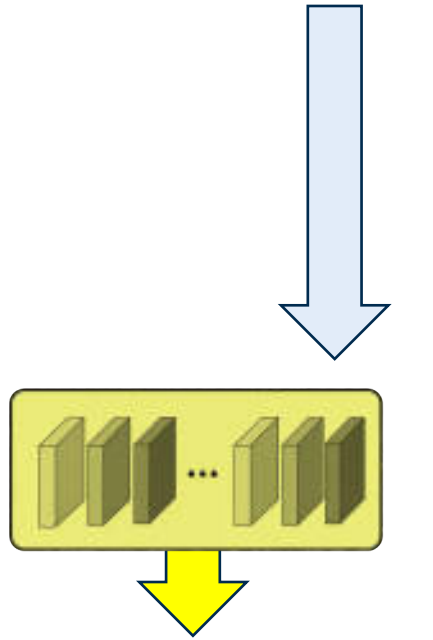


Occupancy Networks

Idea:

- Instead of storing a voxel grid, or a point cloud …
- … can we train a network to predict if a point (x,y,z) is occupied or not?

- Immediate downside: have to learn a new network for each object / scene to be represented.
- Can we add something to re-use the network for many different objects?

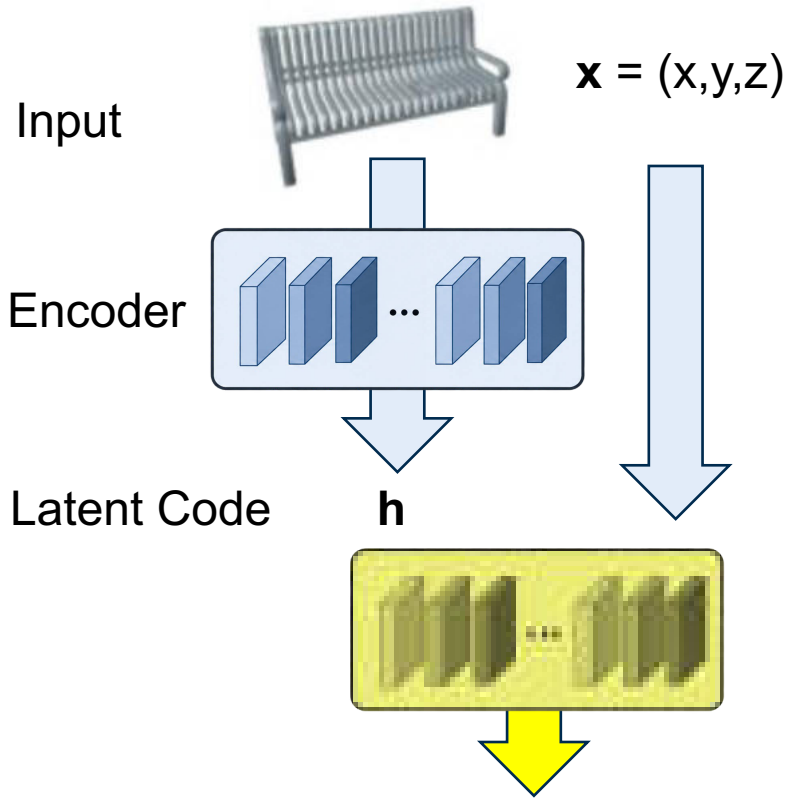


Occupancy Networks

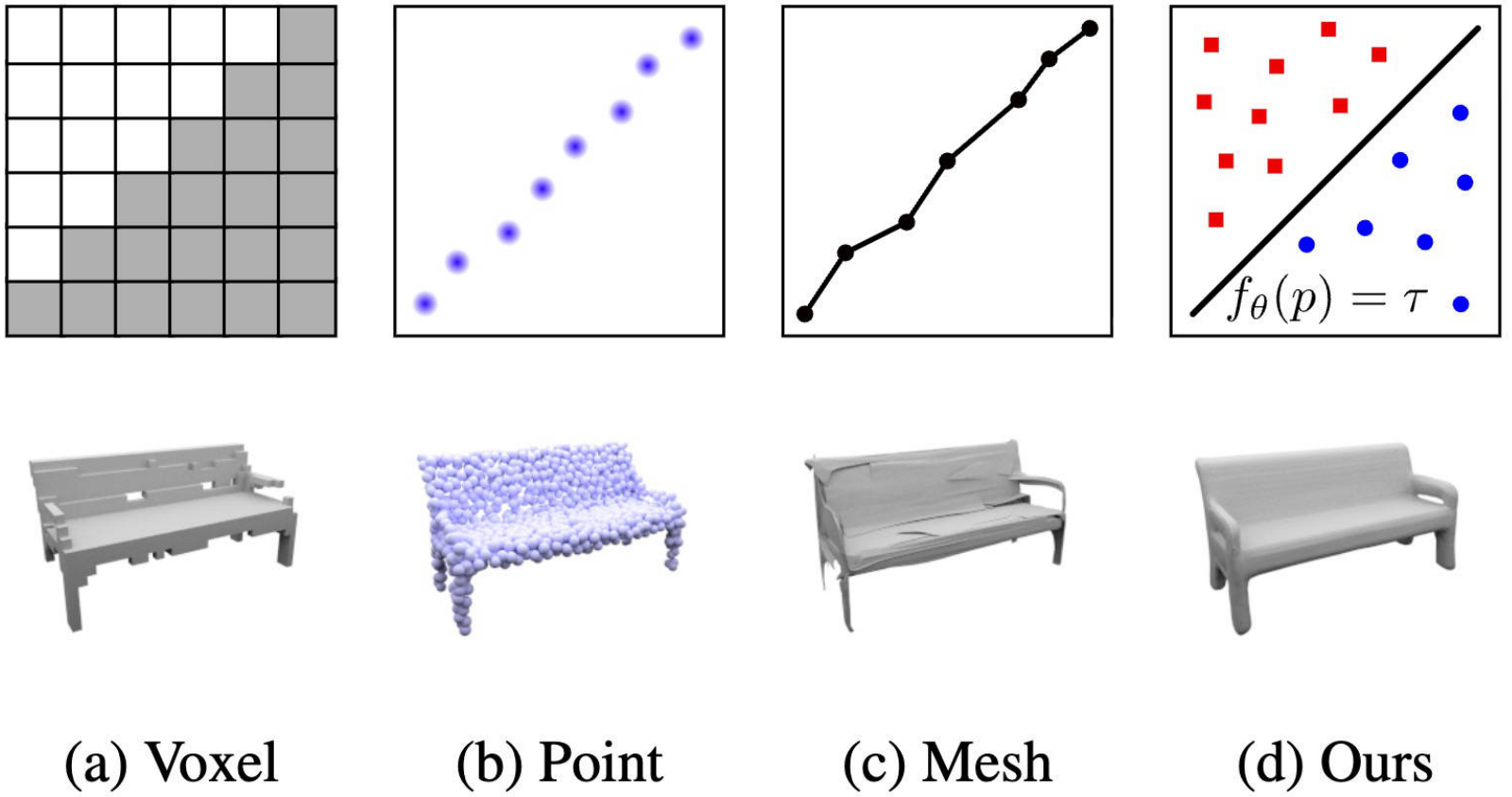
- Represent 3D geometry as a learned **continuous function that maps 3D points to occupancy**
- Input observation (image, point cloud, voxels) is encoded into a latent feature **h**
- For any 3D query point the network predicts

$$f_\theta(\mathbf{x}, \mathbf{h}) \rightarrow p(\text{occupied})$$

- The surface is the decision boundary, e.g. $p(\text{occupied}) = 0.5$
- Unlike a voxel grid, the representation is continuous and can be queried at arbitrary spatial resolution
- Training uses sampled 3D points with inside / outside occupancy labels



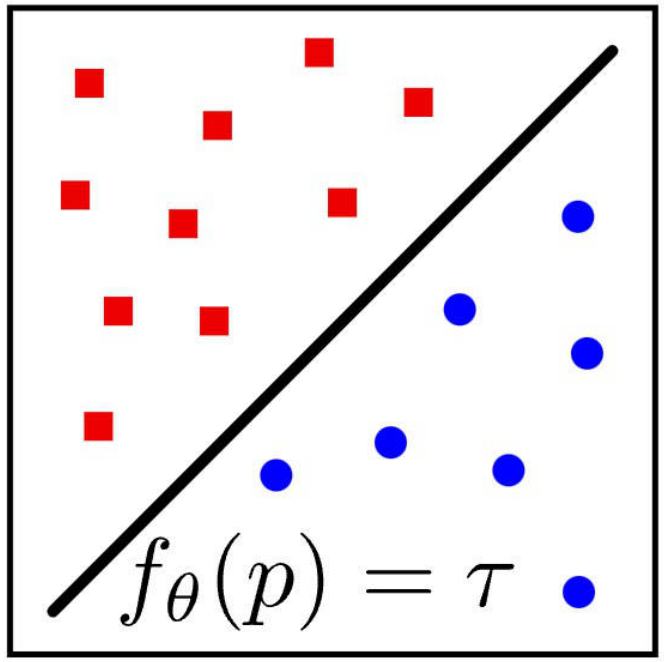
Occupancy Networks

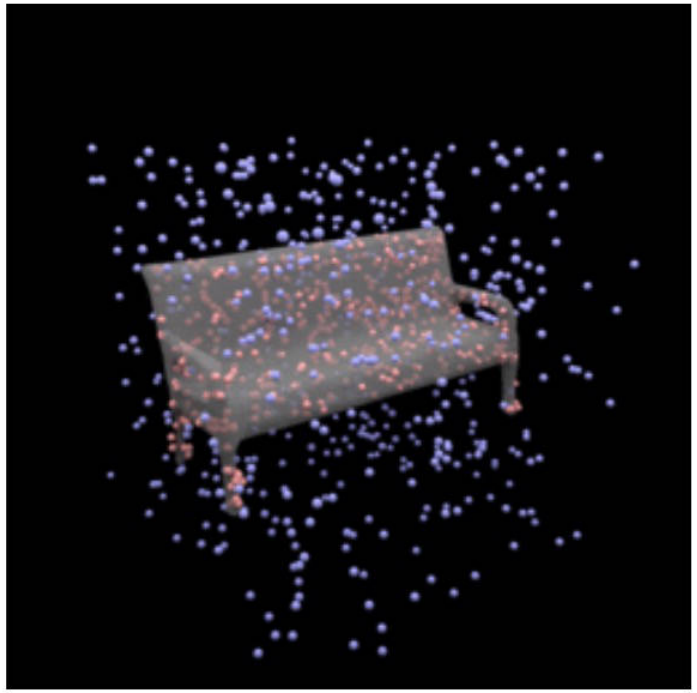


CRICOS No.00213J

Occupancy Networks: Learning 3D Reconstruction in Function Space, Mescheder et al. (2019)

Occupancy Networks





CRICOS No.00213J

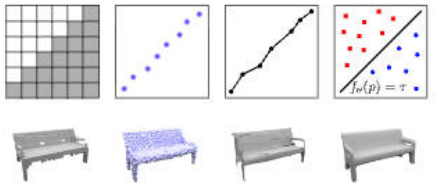
Occupancy Networks: Learning 3D Reconstruction in Function Space

Lars Mescheder[1] Michael Oechsle[1,2] Michael Niemeyer[1] Sebastian Nowozin[3†] Andreas Geiger[1]
[1]Autonomous Vision Group, MPI for Intelligent Systems and University of Tübingen
[2]ETAS GmbH, Stuttgart
[3]Google AI Berlin
{firstname.lastname}@tue.mpg.de nowozin@gmail.com

arXiv:1812.03828v2 [cs.CV] 30 Apr 2019

Abstract

With the advent of deep neural networks, learning-based approaches for 3D reconstruction have gained popularity. However, unlike for images, in 3D there is no canonical representation which is both computationally and memory efficient yet allows for representing high-resolution geometry of arbitrary topology. Many of the state-of-the-art learning-based 3D reconstruction approaches can hence only represent very coarse 3D geometry or are limited to a restricted domain. In this paper, we propose Occupancy Networks, a new representation for learning-based 3D reconstruction methods. Occupancy networks implicitly represent the 3D surface as the continuous decision boundary of a deep neural network classifier. In contrast to existing approaches, our representation encodes a description of the 3D output at infinite resolution without excessive memory footprint. We validate that our representation can efficiently encode 3D structure and can be inferred from various kinds of input. Our experiments demonstrate competitive results, both qualitatively and quantitatively, for the challenging tasks of 3D reconstruction from single images, noisy point clouds and coarse discrete voxel grids. We believe that occupancy networks will become a useful tool in a wide variety of learning-based 3D tasks.



(a) Voxel (b) Point (c) Mesh (d) Ours

Figure 1: **Overview:** Existing 3D representations discretize the output space differently: (a) spatially in voxel representations, (b) in terms of predicted points, and (c) in terms of vertices for mesh representations. In contrast, (d) we propose to consider the continuous decision boundary of a classifier f_θ (e.g., a deep neural network) as a 3D surface which allows to extract 3D meshes at any resolution.

1. Introduction

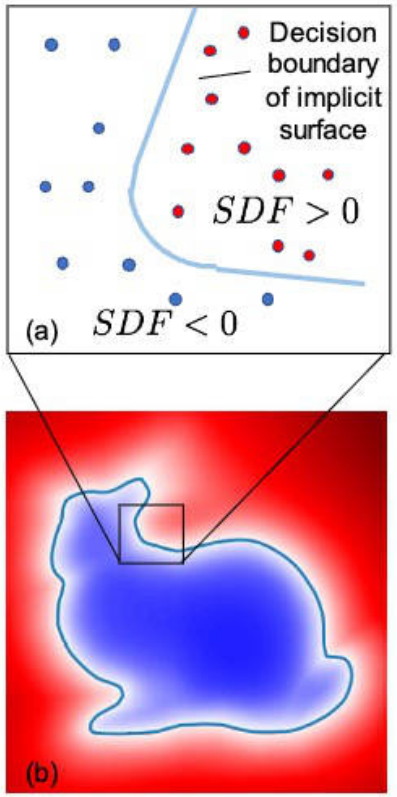
Recently, learning-based approaches for 3D reconstruction have gained popularity [4, 9, 23, 58, 75, 77]. In contrast to traditional multi-view stereo algorithms, learned models are able to encode rich prior information about the space of 3D shapes which helps to resolve ambiguities in the input.

While generative models have recently achieved remarkable successes in generating realistic high resolution images [36, 47, 72], this success has not yet been replicated in the 3D domain. In contrast to the 2D domain, the community has not yet agreed on a 3D output representation that is both memory efficient and can be efficiently inferred from data. Existing representations can be broadly categorized into three categories: voxel-based representations [4, 19, 43, 58, 64, 69, 75], point-based representations [1, 17] and mesh representations [34, 57, 70], see Fig. 1.

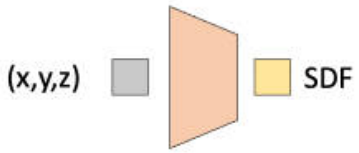
Voxel representations are a straightforward generalization of pixels to the 3D case. Unfortunately, however, the memory footprint of voxel representations grows cubically with resolution, hence limiting naïve implementations to 32^3 or 64^3 voxels. While it is possible to reduce the memory footprint by using data adaptive representations such as octrees [61, 67], this approach leads to complex implementations and existing data-adaptive algorithms are still limited to relatively small 256^3 voxel grids. Point clouds [1, 17] and meshes [34, 57, 70] have been introduced as alternative representations for deep learning, using appropriate loss functions. However, point clouds lack the connectivity structure of the underlying mesh and hence require additional post-processing steps to extract 3D geometry from the model.

[†]Part of this work was done while at MSR Cambridge.

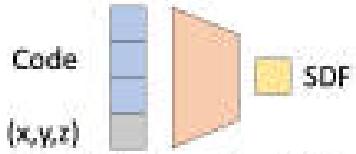
DeepSDF







(a) Single Shape DeepSDF



(b) Coded Shape DeepSDF

arXiv:1901.05103v1 [cs.CV] 16 Jan 2019

DeepSDF: Learning Continuous Signed Distance Functions for Shape Representation

Jeong Joon Park[1,3†] Peter Florence[2,3†] Julian Straub[3] Richard Newcombe[3] Steven Lovegrove[3]

[1]University of Washington [2]Massachusetts Institute of Technology [3]Facebook Reality Labs



Figure 1: DeepSDF represents signed distance functions (SDFs) of shapes via latent code-conditioned feed-forward decoder networks. Above images are raycast renderings of DeepSDF interpolating between two shapes in the learned shape latent space. Best viewed digitally.

Abstract

Computer graphics, 3D computer vision and robotics communities have produced multiple approaches to representing 3D geometry for rendering and reconstruction. These provide trade-offs across fidelity, efficiency and compression capabilities. In this work, we introduce DeepSDF, a learned continuous Signed Distance Function (SDF) representation of a class of shapes that enables high quality shape representation, interpolation and completion from partial and noisy 3D input data. DeepSDF, like its classical counterpart, represents a shape's surface by a continuous volumetric field: the magnitude of a point in the field represents the distance to the surface boundary and the sign indicates whether the region is inside (-) or outside (+) of the shape, hence our representation implicitly encodes a shape's boundary as the zero-level-set of the learned function while explicitly representing the classification of space as being part of the shapes interior or not. While classical SDF's both in analytical or discretized voxel form typically represent the surface of a single shape, DeepSDF can represent an entire class of shapes. Furthermore, we show state-of-the-art performance for learned 3D shape representation and completion while reducing the model size by an order of magnitude compared with previous work.

† Work performed during internship at Facebook Reality Labs.

1. Introduction

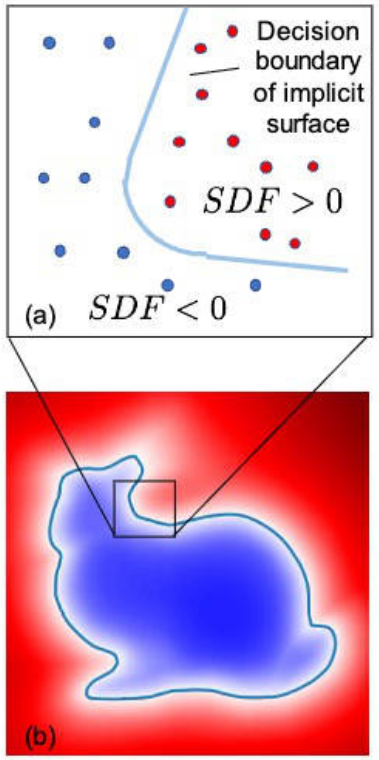
Deep convolutional networks which are a mainstay of image-based approaches grow quickly in space and time complexity when directly generalized to the 3rd spatial dimension, and more classical and compact surface representations such as triangle or quad meshes pose problems in training since we may need to deal with an unknown number of vertices and arbitrary topology. These challenges have limited the quality, flexibility and fidelity of deep learning approaches when attempting to either input 3D data for processing or produce 3D inferences for object segmentation and reconstruction.

In this work, we present a novel representation and approach for generative 3D modeling that is efficient, expressive, and fully continuous. Our approach uses the concept of a SDF, but unlike common surface reconstruction techniques which discretize this SDF into a regular grid for evaluation and measurement denoising, we instead learn a generative model to produce such a continuous field.

The proposed continuous representation may be intuitively understood as a learned shape-conditioned classifier for which the decision boundary is the surface of the shape itself, as shown in Fig. 2. Our approach shares the generative aspect of other works seeking to map a latent space to a distribution of complex shapes in 3D [54], but critically differs in the central representation. While the notion of an

DeepSDF

- Represent 3D geometry as a learned **continuous function that maps 3D points to distance from the nearest surface (= signed distance function)**
- For a 3D query point $\boldsymbol{x} = (x, y, z)$ predict $f_\theta(\boldsymbol{x}, \boldsymbol{h}) \rightarrow s$
- **h** is a latent code
- s is the signed distance to the nearest surface
  - s<0: inside, s=0 on the surface, s>0 outside
- Compared with occupancy, SDF gives a richer signal: not just inside/outside, but how far from the surface



(c)



CRICOS No.00213J

Summary – Learned Implicit Geometry

A neural network can itself be the 3D representation of an object or a scene

- Occupancy Networks:
  learn a continuous function that answers “is this 3D point inside/occupied?”
- DeepSDF:
  learn a continuous function that answers “how far is this 3D point from the surface?”

Next natural question:
If a network can represent geometry,
can it also represent what the scene looks like?

Learn Geometry and Appearance

If a neural field can represent geometry, it can also represent what the scene looks like.

What is the next step?

- What else can we predict?
  - DUSt3r: Image pairs → 3D geometry as point maps
  - Occupancy Networks: (x,y,z) → occupancy
  - DeepSDFs: (x,y,z) → signed distance to nearest surface

- But we could also predict colour, semantic class, learned feature embeddings …

Neural Radiance Fields (NeRFs)

- DUSt3R: images → predict 3D geometry directly
  - Needs ground truth point maps to train!

- NeRF:
  - Does not need ground truth 3D geometry to train!
  - Instead, optimise the scene representation so that it renders the image correctly
  - The representation is in the network weights (in that sense similar to occupancy networks)



arXiv:2003.08934v2 [cs.CV] 3 Aug 2020

NeRF: Representing Scenes as Neural Radiance Fields for View Synthesis

Ben Mildenhall[1]* Pratul P. Srinivasan[1]* Matthew Tancik[1]*
Jonathan T. Barron[2] Ravi Ramamoorthi[3] Ren Ng[1]

[1]UC Berkeley [2]Google Research [3]UC San Diego

Abstract. We present a method that achieves state-of-the-art results for synthesizing novel views of complex scenes by optimizing an underlying continuous volumetric scene function using a sparse set of input views. Our algorithm represents a scene using a fully-connected (non-convolutional) deep network, whose input is a single continuous 5D coordinate (spatial location (x, y, z) and viewing direction (θ, ϕ)) and whose output is the volume density and view-dependent emitted radiance at that spatial location. We synthesize views by querying 5D coordinates along camera rays and use classic volume rendering techniques to project the output colors and densities into an image. Because volume rendering is naturally differentiable, the only input required to optimize our representation is a set of images with known camera poses. We describe how to effectively optimize neural radiance fields to render photorealistic novel views of scenes with complicated geometry and appearance, and demonstrate results that outperform prior work on neural rendering and view synthesis. View synthesis results are best viewed as videos, so we urge readers to view our supplementary video for convincing comparisons.

Keywords: scene representation, view synthesis, image-based rendering, volume rendering, 3D deep learning

1 Introduction

In this work, we address the long-standing problem of view synthesis in a new way by directly optimizing parameters of a continuous 5D scene representation to minimize the error of rendering a set of captured images.

We represent a static scene as a continuous 5D function that outputs the radiance emitted in each direction (θ, ϕ) at each point (x, y, z) in space, and a density at each point which acts like a differential opacity controlling how much radiance is accumulated by a ray passing through (x, y, z). Our method optimizes a deep fully-connected neural network without any convolutional layers (often referred to as a multilayer perceptron or MLP) to represent this function by regressing from a single 5D coordinate (x, y, z, θ, ϕ) to a single volume density and view-dependent RGB color. To render this *neural radiance field* (NeRF)

* Authors contributed equally to this work.





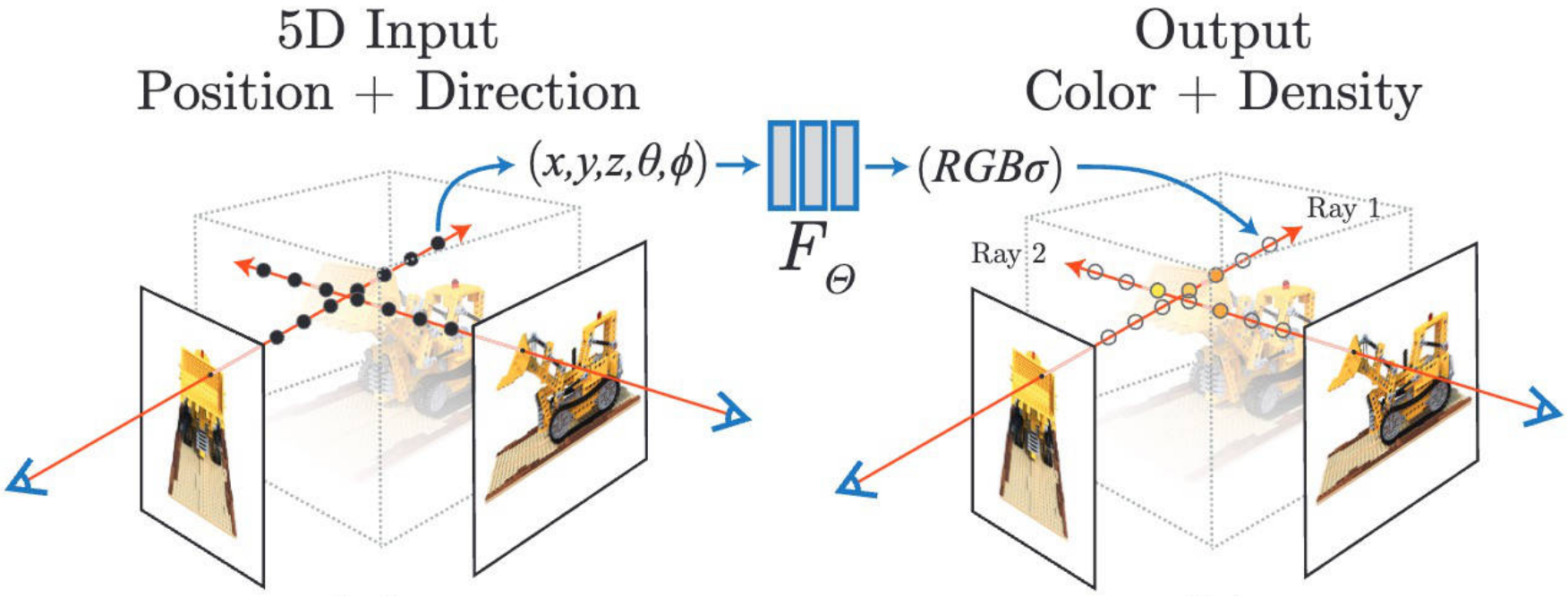








Neural Radiance Fields (NeRFs)

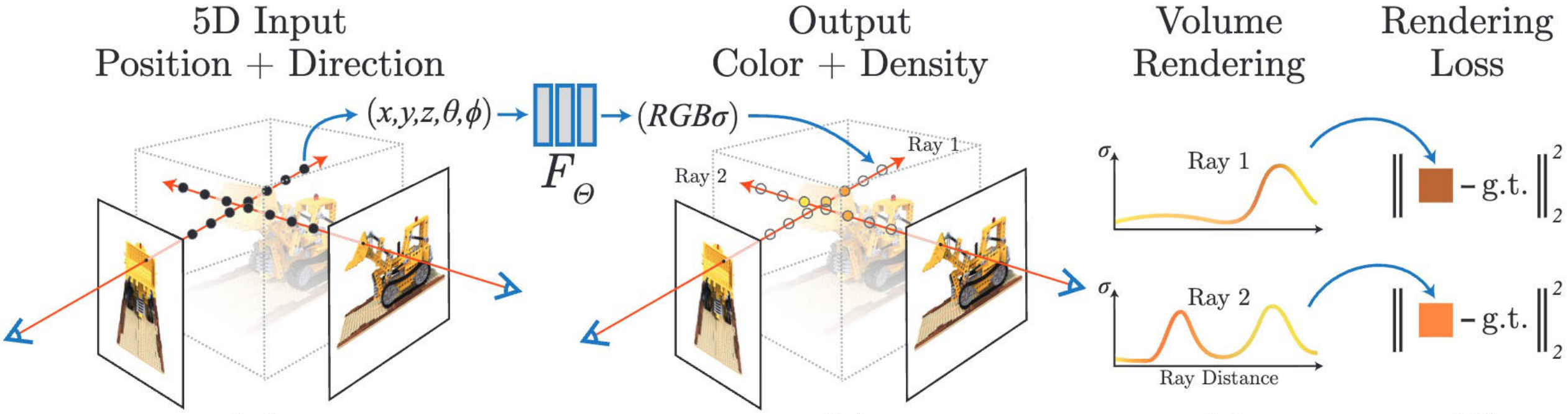


$$f_\theta(x, y, z, \theta, \phi) \rightarrow (\sigma, \mathrm{R}, \mathrm{G}, \mathrm{B})$$

A learned function that maps 3D point + viewing direction to density and colour.

CRICOS No.00213J

Neural Radiance Fields (NeRFs)



$$f_\theta(x, y, z, \theta, \phi) \rightarrow (\sigma, \mathrm{R}, \mathrm{G}, \mathrm{B})$$

A learned function that maps 3D point + viewing direction to density and colour.

CRICOS No.00213J

Neural Radiance Fields (NeRFs)

- Represent the scene as a continuous neural function:
$$f_\theta(x, y, z, \theta, \phi) \rightarrow (\sigma, \mathrm{R}, \mathrm{G}, \mathrm{B})$$
- Input: 3D position (x, y, z) and viewing directions (θ, ϕ)
- Output:
  - Density σ: explains where scene content exists
  - Colour $(\mathrm{R}, \mathrm{G}, \mathrm{B})$: how that point appears from direction (θ, ϕ)
- Geometry is encoded implicitly in the learned density field
- Appearance can be view-dependent
- Key idea: NeRF extends implicit geometry from “is there a surface here?” to “what is here, and what does it look like?”



arXiv:2003.08934v2 [cs.CV] 3 Aug 2020

NeRF: Representing Scenes as Neural Radiance Fields for View Synthesis

Ben Mildenhall[1⋆] Pratul P. Srinivasan[1⋆] Matthew Tancik[1⋆]
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⋆ Authors contributed equally to this work.

Neural Radiance Fields (NeRFs)

For training:

- Start with multiple posed images
- For each training pixel, cast a ray into the scene and sample many 3D points along the ray.
- Query the NeRF at each sample:

$$f_\theta(x, y, z, \theta, \phi) \rightarrow (\sigma, \mathrm{R}, \mathrm{G}, \mathrm{B})$$

- Combine densities and colours along each ray using differentiable volume rendering
- This produces a predicted pixel colour, compare it with the true colour.
- The difference is the loss for training



arXiv:2003.08934v2 [cs.CV] 3 Aug 2020

NeRF: Representing Scenes as Neural Radiance Fields for View Synthesis

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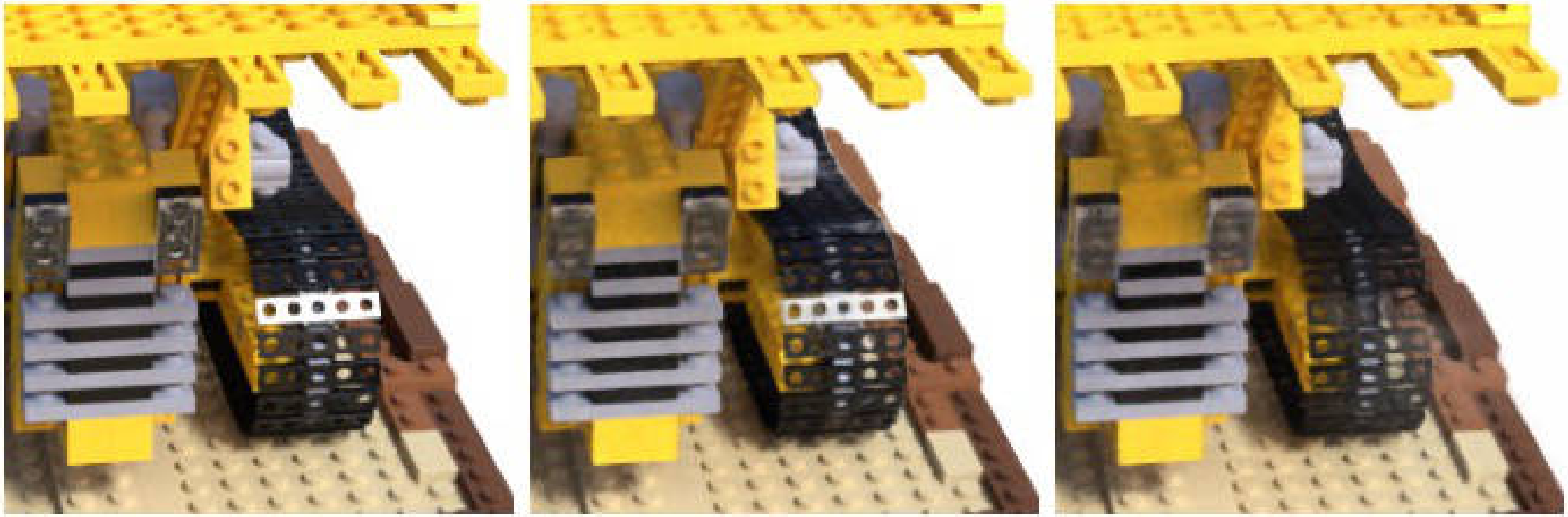
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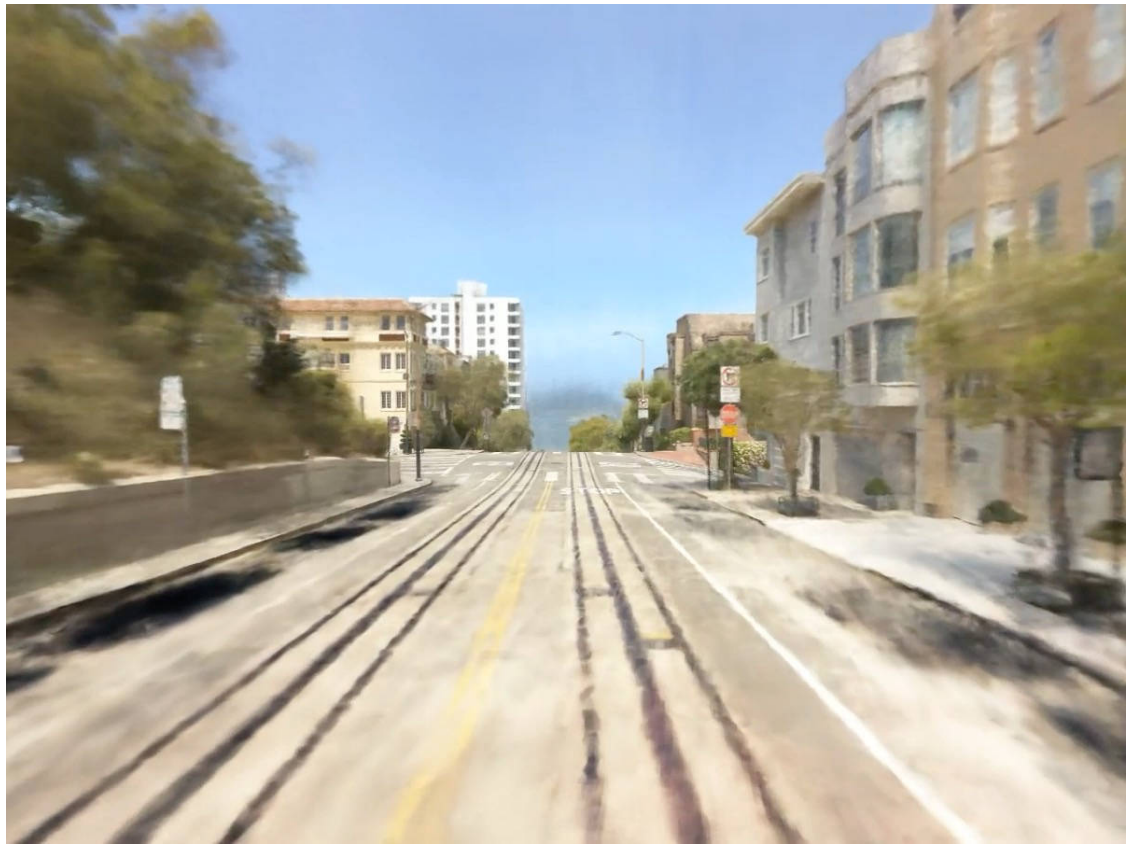
Neural Radiance Fields (NeRFs)

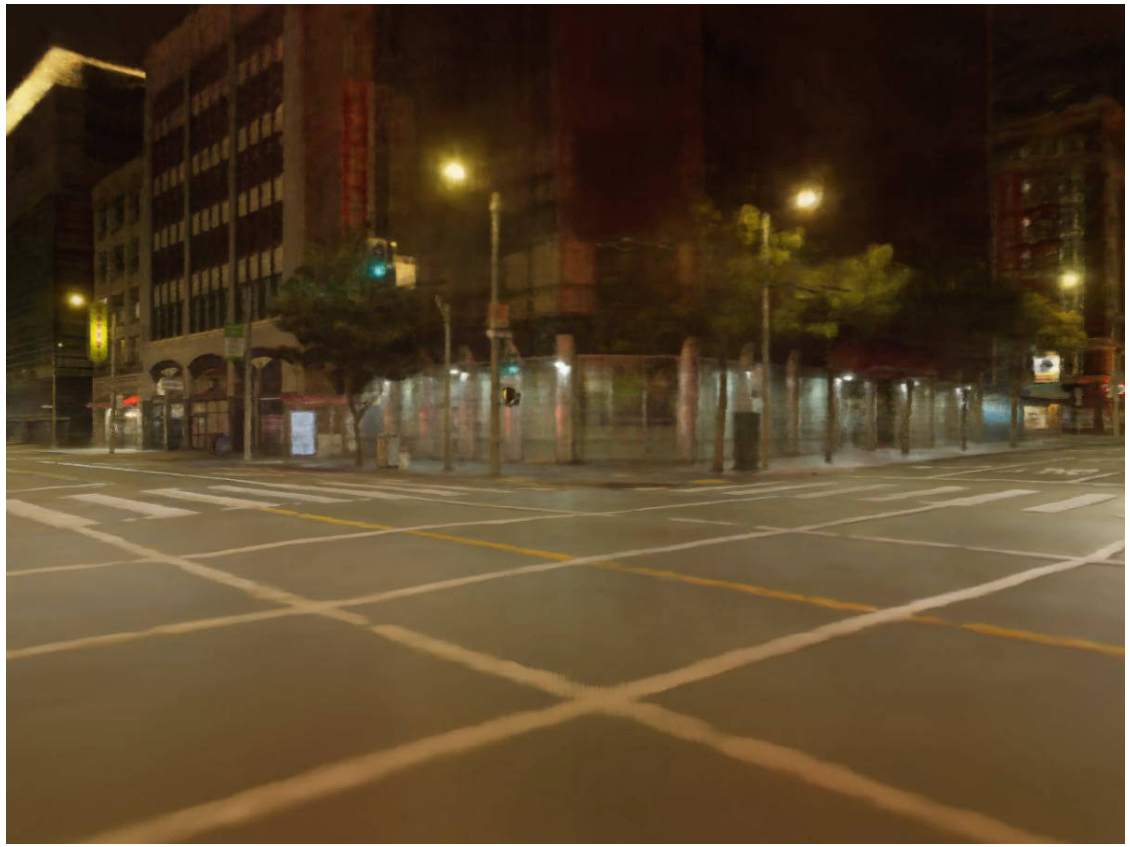


Ground Truth | Complete Model | No View Dependence

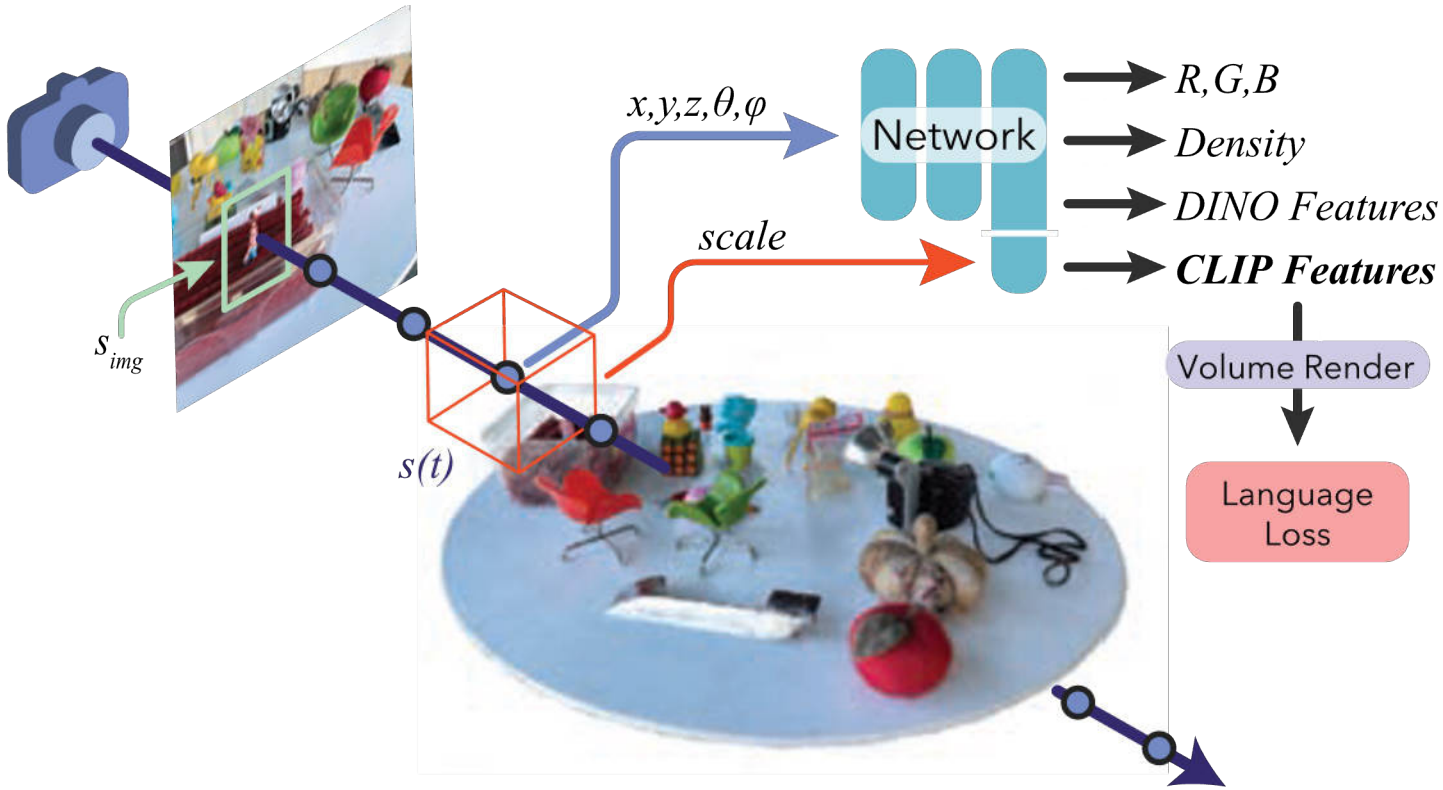
Large-Scale Mapping with NeRFs

Videos courtesy of Block-NeRF (waymo.com/research/block-nerf/)





LERF = NeRF + CLIP



arXiv:2303.09553v1 [cs.CV] 16 Mar 2023

LERF: Language Embedded Radiance Fields

Justin Kerr*, Chung Min Kim*, Ken Goldberg, Angjoo Kanazawa, and Matthew Tancik

UC Berkeley

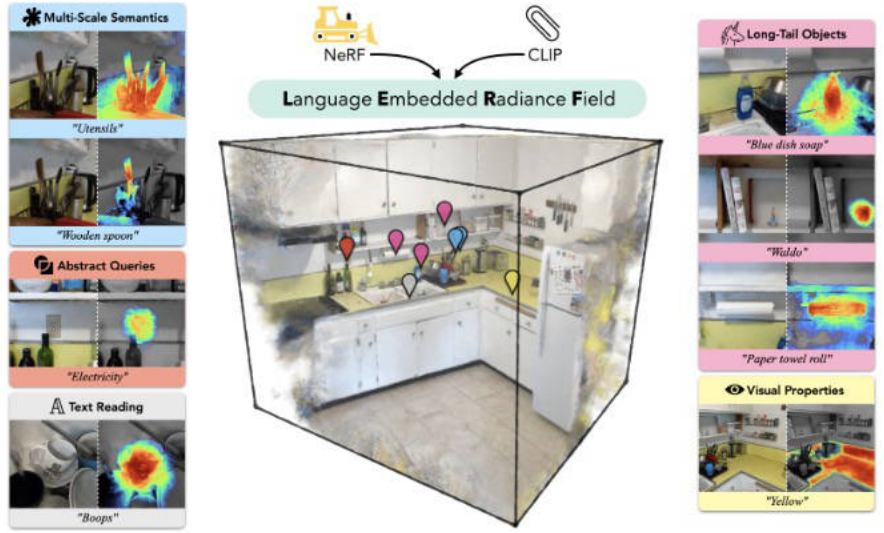


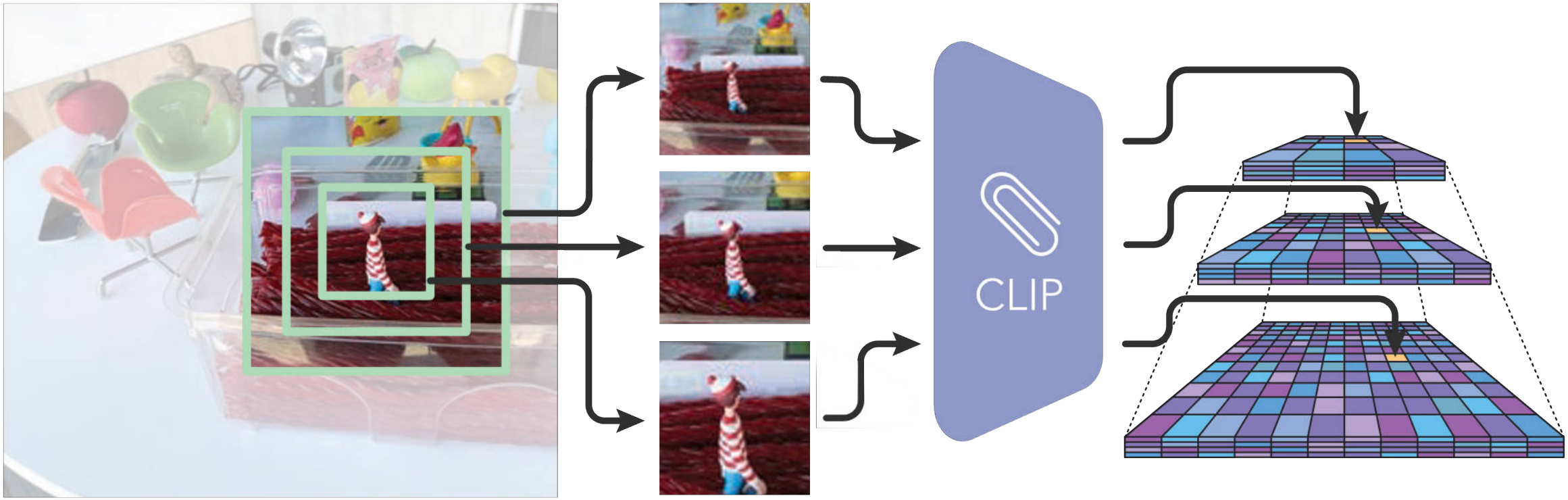
Figure 1: **Language Embedded Radiance Fields (LERF).** LERF grounds CLIP representations in a dense, multi-scale 3D field. A LERF can be reconstructed from a hand-held phone capture within 45 minutes, then can render dense relevancy maps given textual queries interactively in real-time. LERF enables a broad range of concepts to be queried via natural language, from abstract queries like *"Electricity"*, visual properties like *"Yellow"*, long-tail objects such as *"Waldo"*, and even reading text like *"Boops"* on the mug. For each prompt, an RGB image and relevancy map are rendered focusing on the location with maximum relevancy activation.

Abstract

Humans describe the physical world using natural language to refer to specific 3D locations based on a vast range of properties: visual appearance, semantics, abstract associations, or actionable affordances. In this work we propose Language Embedded Radiance Fields (LERFs), a method for grounding language embeddings from off-the-shelf models like CLIP into NeRF, which enable these types of open-ended language queries in 3D. LERF learns a dense, multi-scale language field inside NeRF by volume rendering CLIP embeddings along training rays, supervising these embeddings across training views to provide multi-view consistency and smooth the underlying language field. After optimization, LERF can extract 3D relevancy maps for a broad range of language prompts interactively in real-time, which has potential use cases in robotics, understanding vision-language models, and interacting with 3D scenes. LERF enables pixel-aligned, zero-shot queries on the distilled 3D

*Equal contribution, corresponding authors.

LERF = NeRF + CLIP



CRICOS No.00213J



Neural Radiance Fields – Summary

Crucial Point:
NeRF does not need ground-truth 3D geometry!
Geometry emerges because the field must explain all input views consistently.

Next Question:
NeRF stores the scene **implicitly inside a neural network**.
Could we instead store **learnable 3D primitives explicitly**
and render them much faster?

CRICOS No.00213J

Explicit Learned Scene Representation

Do we have to represent the scene implicitly inside the parameters of a network?

3D Gaussian Splatting

- NeRF gave us the idea of learning geometry + appearance jointly from images.
- But do we actually need to store the scene inside a network? 3DGS says: **no.**

- NeRF: scene representation is implicit (in the network weights)
- 3DGS scene is explicit (represented by 3D Gaussian Ellipsoids)
- Both don't need ground truth geometry, but learn by comparing the rendered images to the training images



3D Gaussian Splatting for Real-Time Radiance Field Rendering

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Fig. 1. Our method achieves real-time rendering of radiance fields with quality that equals the previous method with the best quality [Barron et al. 2022], while only requiring optimization times competitive with the fastest previous methods [Fridovich-Keil and Yu et al. 2022; Müller et al. 2022]. Key to this performance is a novel 3D Gaussian scene representation coupled with a real-time differentiable renderer, which offers significant speedup to both scene optimization and novel view synthesis. Note that for comparable training times to InstantNGP [Müller et al. 2022], we achieve similar quality to theirs; while this is the maximum quality they reach, by training for 51min we achieve state-of-the-art quality, even slightly better than Mip-NeRF360 [Barron et al. 2022].

Radiance Field methods have recently revolutionized novel-view synthesis of scenes captured with multiple photos or videos. However, achieving high visual quality still requires neural networks that are costly to train and render, while recent faster methods inevitably trade off speed for quality. For unbounded and complete scenes (rather than isolated objects) and 1080p resolution rendering, no current method can achieve real-time display rates. We introduce three key elements that allow us to achieve state-of-the-art visual quality while maintaining competitive training times and importantly allow high-quality real-time (≥ 30 fps) novel-view synthesis at 1080p resolution. First, starting from sparse points produced during camera calibration, we represent the scene with 3D Gaussians that preserve desirable properties of continuous volumetric radiance fields for scene optimization while avoiding unnecessary computation in empty space; Second, we perform interleaved optimization/density control of the 3D Gaussians, notably optimizing anisotropic covariance to achieve an accurate representation of the scene; Third, we develop a fast visibility-aware rendering algorithm that supports anisotropic splatting and both accelerates training and allows real-time rendering. We demonstrate state-of-the-art visual quality and real-time rendering on several established datasets.

CCS Concepts: • **Computing methodologies → Rendering**; **Point-based models**; **Rasterization**; **Machine learning approaches**.

Additional Key Words and Phrases: novel view synthesis, radiance fields, 3D gaussians, real-time rendering

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1 INTRODUCTION

Meshes and points are the most common 3D scene representations because they are explicit and are a good fit for fast GPU/CUDA-based rasterization. In contrast, recent Neural Radiance Field (NeRF) methods build on continuous scene representations, typically optimizing a Multi-Layer Perceptron (MLP) using volumetric ray-marching for novel-view synthesis of captured scenes. Similarly, the most efficient radiance field solutions to date build on continuous representations by interpolating values stored in, e.g., voxel [Fridovich-Keil and Yu et al. 2022] or hash [Müller et al. 2022] grids or points [Xu et al. 2022]. While the continuous nature of these methods helps optimization, the stochastic sampling required for rendering is costly and can result in noise. We introduce a new approach that combines the best of both worlds: our 3D Gaussian representation allows optimization with state-of-the-art (SOTA) visual quality and competitive training times, while our tile-based splatting solution ensures real-time rendering at SOTA quality for 1080p resolution on several previously published datasets [Barron et al. 2022; Hedman et al. 2018; Knapitsch et al. 2017] (see Fig. 1).

Our goal is to allow real-time rendering for scenes captured with multiple photos, and create the representations with optimization times as fast as the most efficient previous methods for typical real scenes. Recent methods achieve fast training [Fridovich-Keil

3D Gaussian Splatting



Each Gaussian carries:

- Centre and shape
  i.e. mean + covariance
- Opacity
- Direction-dependent colour
  (as spherical harmonics)

Illustration by GPT5.6

3D Gaussian Splatting

Each 3D Gaussian projects to a soft 2D ellipse on the image plane. Rendering means “splatting” all of those ellipses onto the image and blending them according to depth and opacity.

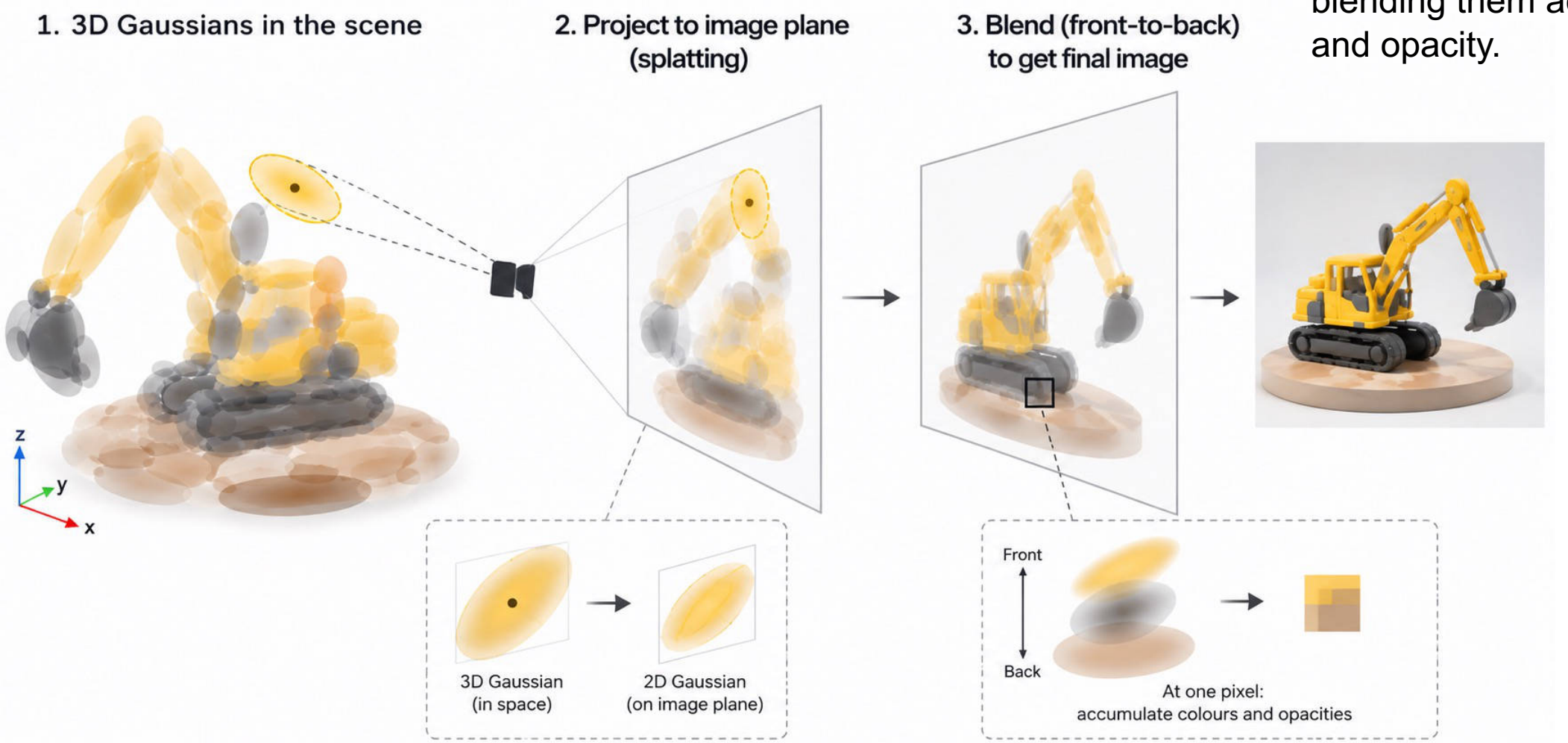
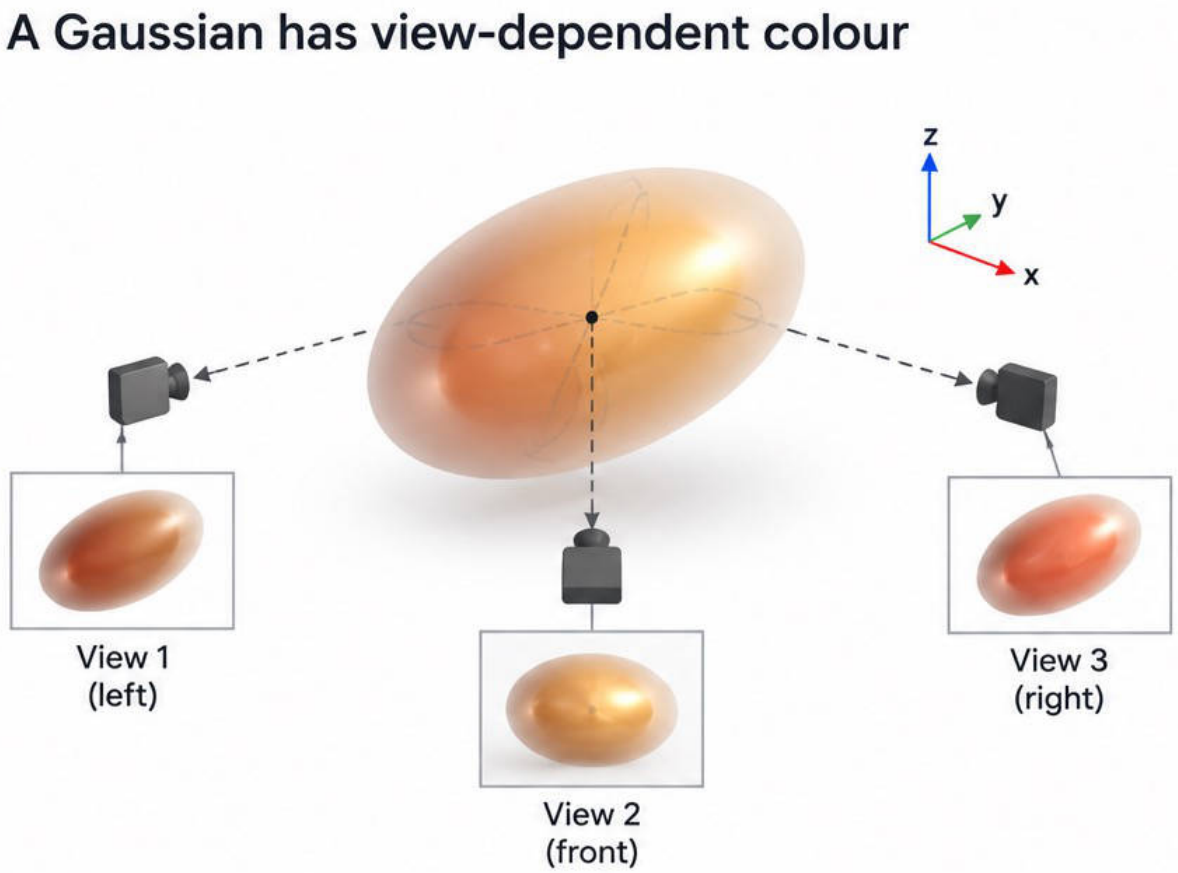


Illustration by GPT5.6

3D Gaussian Splatting – Spherical Harmonics Encode View-Dependent Colour



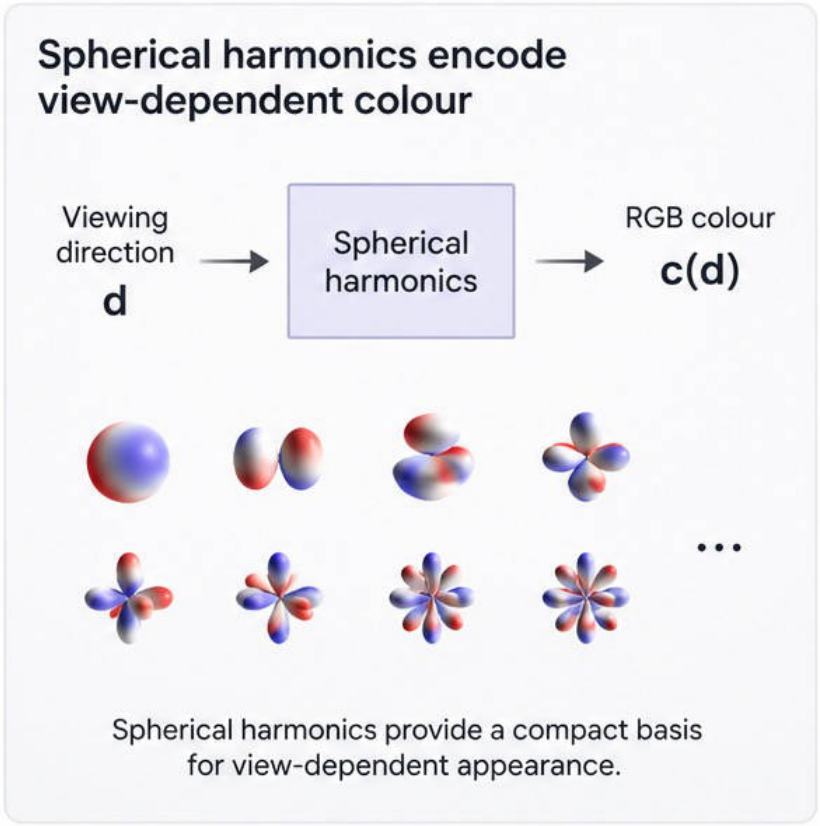


Illustration by GPT5.6



This video contains a voice-over

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(ACM Transactions on Graphics)

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GraphDeco
GRAPHics and Design with Heterogeneous COntent

LangSplat = 3DGS + CLIP



LangSplat: 3D Language Gaussian Splatting

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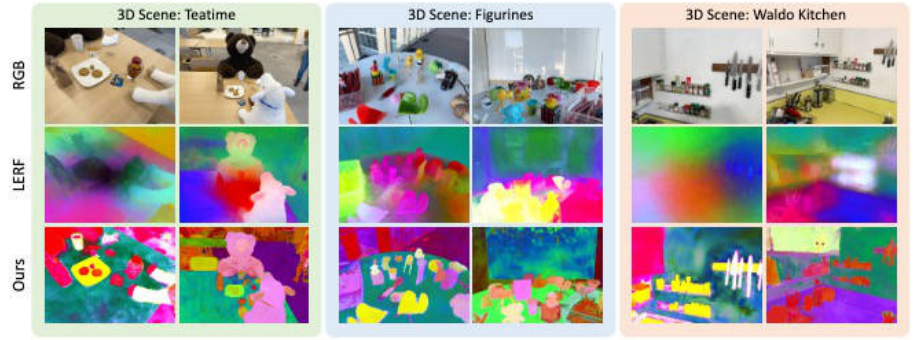


Figure 1. Visualization of learned 3D language features of the previous SOTA method LERF and our LangSplat. While LERF generates imprecise and vague 3D features, our LangSplat accurately captures object boundaries and provides precise 3D language fields. While being effective, our LangSplat is also **199** × faster than LERF at the resolution of 1440 × 1080.

Abstract

Humans live in a 3D world and commonly use natural language to interact with a 3D scene. Modeling a 3D language field to support open-ended language queries in 3D has gained increasing attention recently. This paper introduces LangSplat, which constructs a 3D language field that enables precise and efficient open-vocabulary querying within 3D spaces. Unlike existing methods that ground CLIP language embeddings in a NeRF model, LangSplat advances the field by utilizing a collection of 3D Gaussians, each encoding language features distilled from CLIP, to represent the language field. By employing a tile-based splatting technique for rendering language features, we circumvent the costly rendering process inherent in NeRF. Instead of directly learning CLIP embeddings, LangSplat first trains a scene-wise language autoencoder and then learns language features on the scene-specific latent space, thereby alleviating substantial memory demands imposed by explicit modeling. Existing methods struggle with imprecise and vague 3D language fields, which fail to discern clear boundaries between objects. We delve into this issue and propose to learn hierarchical semantics using SAM, thereby eliminating the need for extensively querying the language field across various scales and the regularization of DINO features. Extensive experimental results show that LangSplat significantly outperforms the previous state-of-the-art method LERF by a large margin. Notably, LangSplat is extremely efficient, achieving a 199 × speedup compared to LERF at the resolution of 1440 × 1080. We strongly recommend readers to check out our video results at https://langsplat.github.io/.

1. Introduction

Language is the primary means of communication for human beings [3]. Modeling a 3D language field allows users

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Summary

From Classical to Learned Geometry

- Earlier in the semester, we built 3D understanding through an **explicit pipeline**: **features** → **matches** → **geometry** → **pose** → **3D**
- Classical methods explicitly compute geometry; modern methods increasingly **learn to predict geometrically useful quantities** directly.

- The geometry has not disappeared: it is increasingly embedded inside learned representations and training objectives:
  - Monocular depth: image → depth
  - DUSt3R: image pair → dense 3D point maps
  - MASt3R: point maps + accurate learned correspondences

CRICOS No.00213J

From *"Where is stuff?"* and *"What does it look like?"* to *"What is it?"*

How to represent the 3D world?

- Point clouds: explicit surface samples
- Voxels / occupancy grids: discrete free/occupied space
- Meshes: explicit connected surfaces

- Occupancy Networks / DeepSDF: continuous learned geometry
- NeRF: implicit geometry + appearance
- 3D Gaussian Splatting: explicit learnable primitives + fast rendering
- LERF / LangSplat: geometry + appearance + semantic/language features

What is the Best Representation?

- The “best” representation depends on the task:
  - VO/SLAM → geometry and consistency
  - planning → free/occupied space
  - rendering → appearance
  - semantic robotics → language-aligned 3D understanding

CRICOS No.00213J

Modern robotic vision is moving from hand-designed geometric pipelines toward **learned models** that jointly recover and represent **geometry and semantics** of 3D scenes.

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